Cryogenics 143 (2024) 103950

Contents lists available at ScienceDirect

Cryogenics

journal homepage: www.elsevier.com/locate/cryogenics

Progress in measuring techniques and thermal radiative properties of metals at cryogenic temperatures: A review

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ARTICLE INFO

Keywords:
Emissivity measurement
Absorptivity
Cryogenics
Calorimetric
Radiometric
Metals

ABSTRACT

In recent years, the development in measurement techniques of thermal radiative properties has attracted much attention to cryogenic applications, including space exploration, cryogenic research, medical applications, instrument design, planetary exploration and remote sensing, cryogenic storage and transportation. This paper discusses the importance of emissivity / absorptivity measurement, the factors they depend on, and the respective measurement methods, including calorimetric and radiometric. This paper also summarizes the published data on emissivity and absorptivity related to aluminum and stainless steel. It highlights the importance of surface treatment of materials, providing valuable insight into the significance of emissivity in cryogenic applications, which can serve as a reference for future research in this field.

1. Introduction

In cryogenics, where temperatures drop to extremely low levels precise and comprehensive measurements are paramount to mitigating processes associated with that low temperature. The field of Cryogenics deals with the behavior of a material at extremely low temperature, and it has witnessed remarkable advancement in recent years, opening new frontiers in science and technology. One critical aspect at the forefront of this exploration is understanding and measuring emissivity / absorptivity at cryogenic temperatures.

The criteria for including papers in this article depended on reviewing all research papers on emissivity and absorptivity measurement published in journals such as the International Journal of Refrigeration and Cryogenics. The article also examined proceedings such as Advances in Cryogenic Engineering and carefully selected papers that provide methods for measuring emissivity or absorptivity at cryogenic temperatures. Additionally, databases were considered that specifically published data for the emissivity and absorptivity of aluminum and stainless steel only.

1.1. Importance of emissivity at cryogenic temperature

The study of cryogenic thermal dynamics involves investigating the intricacies of molecular motion. Understanding this field requires careful examination of molecular behavior at extremely low temperatures. Translation and rotation take center stage at lower energy levels, but an uptick in energy triggers a shift towards molecular vibration and more frequent collisions. This transition ultimately results in thermal radiation taking on a more prominent role as the primary energy dissipation mode. Kirchhoff's law serves as a fundamental principle, outlining how thermal radiation relies on various factors, including material properties such as temperature, absorptivity, and emissivity. Thermal energy transfer mechanisms at cryogenic temperatures, without convective currents, include heat conduction through solid materials, gas conduction, and thermal radiation [1].

All the above components that contribute to thermal transfer must be reduced for a system operating at cryogenic temperatures. Solid heat conduction can be reduced by carefully handling the structure's geometry and selecting appropriate materials. Heat transfer through conduction in gases could result from collision between molecules and the surface of the container. The Knudson Number can affect this transfer at lower pressure. At higher Knudson numbers, heat transfer depends purely on the number of molecules and changes linearly with pressure. When the pressure falls below 0.01 Pa, gas heat conduction can be minimized, and radiation heat transfer becomes more significant [2,3].

Emissivity plays a critical role in minimizing radiative heat transfer in cryogenic systems, where other modes of heat transfer, like conduction and convection, are significantly reduced. Low-emissivity

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https://doi.org/10.1016/j.cryogenics.2024.103950
Received 12 March 2024; Received in revised form 27 August 2024; Accepted 17 September 2024
Available online 19 September 2024
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materials, such as polished aluminum and copper, are essential in cryogenic storage systems to reduce thermal loads and minimize the boil-off rates of cryogenic fluids. Jiang et al. [4] highlighted that controlling emissivity in composite insulation and vapor-cooled shields is crucial for maintaining the efficiency of on-orbit cryogenic storage tanks. Similarly, in superconducting magnet systems, maintaining low temperatures is vital to ensure superconductivity, and the use of low-emissivity materials like copper screens is key to preventing excess heat from compromising system performance. This importance extends to space applications, where Krenn et al. [5] discussed the development of a thermal control coating optimized for cryogenic conditions, emphasizing the need for low-emissivity surfaces to enhance the thermal stability of these systems. Moreover, Youngquist et al. [6] demonstrated that cryogenic deep space missions rely on advanced low-emissivity coatings to effectively manage thermal radiation, ensuring the longevity and efficiency of cryogenic components in space environments.

1.2. Thermal radiation

Understanding the fundamental mechanisms of heat transfer is crucial in studying thermal processes across various scientific and engineering disciplines. Among these mechanisms, thermal radiation is distinguished by its unique property of propagating without needing a medium. This mode of heat transfer is facilitated by the emission of electromagnetic waves from any substance at a temperature above absolute zero, enabling energy exchange even across the vacuum of outer space. The behavior and characteristics of thermal radiation are comprehensively described by fundamental physics laws, foundational in both theoretical and applied research of thermal phenomena.

At the heart of thermal radiation study lies the concept of a black body, an idealized physical entity that absorbs all incident electromagnetic radiation regardless of frequency or angle of incidence and emits energy uniformly in all directions and across all wavelengths. While no real object fully embodies the characteristics of a perfect black body, this concept remains instrumental in thermal radiation analysis, serving as a benchmark for comparing the emissive properties of actual materials.

Fundamental laws establish the theoretical framework for analyzing black body radiation. Planck's Law quantifies the spectral density of radiation emitted by a black body at a specific temperature (T), linking thermal radiation to quantum mechanics.

$$B_\nu = \frac{2h\upsilon^3}{c^2}\frac{1}{\exp\left[\frac{h\upsilon}{kT}\right]-1} \tag{1}$$

Here B_ν represent the spectral radiance, c is the speed of light, υ is the frequency, h is Planck's constant, and k is Stephan's Boltzmann constant [7,8].

Wien's Displacement Law offers insights into the temperature of a black body and its emitted radiation's peak wavelength (λ_{max}), indicating that bodies at higher temperatures emit radiation at shorter wavelengths.

$$\lambda_{max}T = b \tag{2}$$

T is the absolute temperature, and b is Wein's displacement constant [9].

The Stefan-Boltzmann Law demonstrates a direct relationship between the thermal radiation emitted by a body and its temperature, showing that the rate of energy radiation (E) from a surface is proportional to the surface area A, emissivity (ε) and to the fourth power of the absolute (Kelvin) temperature T.

$$E = A\varepsilon\sigma T^4 \tag{3}$$

Here σ is the Stefan-Boltzmann constant [10,11].

Emissivity and absorptivity are interlinked by Kirchhoff's law of thermal radiation, which states that for a body that radiates and absorbs radiation in thermal equilibrium, then the absorptivity is equal to the emissivity [12]

$$\varepsilon(\lambda) = \alpha(\lambda) \tag{4}$$

The ability of a surface to emit and absorb thermal radiation is determined by two significant thermal parameters: emissivity and absorptivity [13].

1.3. Emissivity/absorptivity

Emissivity (ε) and absorptivity are fundamental properties in thermal radiation, crucial for characterizing the radiative behavior of materials. It measures the efficiency of an actual surface in emitting energy as thermal radiation, compared to that of an ideal black body at the same temperature. It is a dimensionless quantity that varies between 0 and 1, with higher values indicating greater emissivity and more efficient energy radiation. Specified radiation from an actual surface, as compared with emission from a blackbody at the same temperature, can be written as

$$\varepsilon = \frac{\textit{radiative heat flow of actual body at temperature } T}{\textit{radiative heat flow of blackbody at temperature } T} \tag{5}$$

Emissivity is categorized into directional spectral emissivity (at a specific wavelength, direction, and temperature), directional total emissivity (integrated over all wavelengths for a particular direction and temperature), hemispherical spectral emissivity (at a specific wavelength and temperature, averaged over all directions), and hemispherical total emissivity (integrated over all wavelengths and directions at a given temperature).

Absorptivity refers to the proportion of energy a body absorbs when exposed to radiation. It's more complex than emissivity because it accounts for the directional and spectral characteristics of the incoming radiation, as well as the temperature of the absorbing surface. Like emissivity, absorptivity values range from 0 to 1. It can be categorized into directional spectral absorptivity, directional total absorptivity, hemispherical spectral absorptivity, and hemispherical total absorptivity, each defined by specific wavelengths, directions, and temperatures.

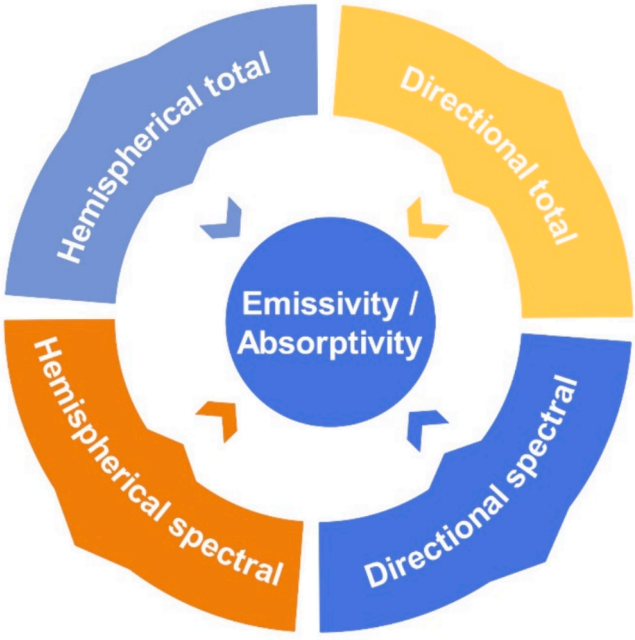


Fig. 1. Classification of emissivity and absorptivity.

The classification of emissivity and absorptivity is depicted in Fig. 1.

The spectral emissive power of a surface (E_λ) is a fundamental concept in radiative heat transfer, describing the power emitted per unit area per unit wavelength. Mathematically, it is expressed as

$$E_\lambda = f(T, \lambda, \theta, \phi) \tag{6}$$

indicating its dependence on the temperature (T), wavelength (λ), and the emission angles θ and ϕ. This function highlights how the emitted radiation varies with temperature, wavelength, and direction. The spectral emissive power increases at higher temperatures, and the peak emission shifts to shorter wavelengths, as Planck's law dictates. Additionally, real surfaces exhibit directional dependence in their emission characteristics, resulting in anisotropic emission patterns. Understanding these dependencies is crucial for accurately modelling and predicting radiative heat transfer in various engineering applications.

1.4. Factors affecting emissivity

Emissivity is influenced by several factors that must be carefully considered to determine a material's heat transfer characteristics accurately. These key factors affect emissivity, including wavelength, surface condition, viewing angle, and temperature.

The wavelength of radiation significantly impacts a material's emissivity, as emissivity varies depending on the wavelength. Surface condition is another crucial factor that impacts emissivity. In the case of metallic materials, the emissivity decreases with polishing and increases with surface roughness and the degree of oxidization. Frolec et al. [14] discussed the effect of surface treatment, such as mechanical and chemical polishing, and roughness on total hemispherical emissivity. Viewing angle is also an essential factor that plays a significant role in determining radiative heat transfer between surfaces, especially at cryogenic temperatures. Recent experimental studies have highlighted the critical impact of viewing angles on the emissivity of materials, particularly at cryogenic temperatures. Gillespie et al. [15] discussed the importance of directional effects in remote sensing, illustrating how varying viewing angles can significantly influence emissivity measurements. Banik et al. [16] further explored this phenomenon in the context of radiative cooling systems, demonstrating that the performance of such systems at cryogenic temperatures is sensitive to changes in viewing angle due to its effect on emissivity. Similarly, Dozier and Warren [17] provided experimental evidence on how the infrared emissivity of snow, used as a proxy for cryogenic materials, varies with viewing angle, underscoring the need for precise angle control in low-temperature applications. Domoto et al. [18] conducted experiments on radiative transfer between metallic surfaces at cryogenic temperatures, confirming that emissivity, and thus the efficiency of thermal management, is highly dependent on the relative viewing angle between surfaces. Finally, temperature is a critical factor that can significantly affect emissivity. As the temperature of a material increases, its emissivity typically increases as well [19–23].

1.5. Applications

Measuring thermal radiative properties at cryogenic temperatures carries a broad spectrum of applications across multiple fields. For instance, in space exploration, emissivity measurement is critical in the James Webb Space Telescope (JWST) to ensure effective thermal control, cryogenic cooling, and accurate infrared observations. The JWST's instruments, especially those designed for infrared detection, require extremely low operating temperatures, and accurate emissivity data are vital for designing thermal systems that maintain these conditions [24]. The telescope's materials and coatings were carefully selected based on their thermal radiative properties to minimize unwanted thermal emissions, which could otherwise interfere with sensitive infrared data [25]. Furthermore, the thermal radiative properties play a key role in the thermal performance of the sun shield, which is crucial for protecting the instruments from solar radiation and ensuring that the telescope's heat does not contaminate observations [26].

Understanding the thermal radiative properties of materials at extremely low temperatures is critical in developing and researching superconductors. Researchers aim to study and analyze the behavior of different materials at such low temperatures in cryogenic research [27]. Thermal radiative properties measurement is used in the thermal shielding of superconducting magnets system in the ITER (International thermonuclear experimental reactor) at 80 K [28]. Superconductors, such as Niobium-tin (Nb_3Sn), exhibit unique electrical and thermal properties that significantly influence their thermal radiative properties. The transition to the superconducting state modifies surface characteristics, potentially impacting interaction with thermal radiation. Additionally, different phases within superconducting materials alter effective thermal radiative properties. Accurate emissivity measurement remains crucial for understanding superconductors' thermal behavior, especially considering temperature-dependent variations [29].

The development and optimization of cryopreservation techniques, which are critical for preserving biological materials such as cells, tissues, or organs for medical applications, also depend on thermal radiative properties. Cryopreservation is a fundamental enabling technology in regenerative medicine, providing a safe and reliable storage method for primary tissue isolates, constructs, and prepared cell preparations over long periods [30].

The Berlin Emissivity Database (BED) is crucial in understanding thermal radiative properties at cryogenic temperatures. When applied to extreme cold conditions, this comprehensive spectral library provides high-quality emissivity spectra within the thermal infrared range (7 – 22 μm). One notable application is its use alongside the MERTIS instrument on the BepiColombo mission to Mercury. By analyzing emissivity at cryogenic temperatures, scientists gain unprecedented insights into the planet's surface and atmosphere, refining our understanding of their composition and physical properties. This knowledge contributes significantly to planetary exploration and remote sensing data interpretation [31,32].

Thermal radiative properties measurement plays a pivotal role in storing and transporting cryogenic fluids, such as liquid hydrogen, by significantly enhancing thermal insulation and safety [33]. By employing materials with low emissivity, radiative heat transfer is minimized, reducing cryogen's boil-off rate and enhancing insulation efficiency. This optimization not only lowers operational costs by decreasing the need for re-liquefaction or additional refrigeration but also contributes to the safety of the storage systems by maintaining stable internal temperatures, thus preventing over-pressurization risks [34]. Furthermore, accurate measurements of thermal radiative properties guide the selection and design of insulation materials, leading to more effective and tailored solutions for cryogenic applications [35,36].

Accurately measuring thermal radiative properties is also important in designing and calibrating infrared cameras in cryogenic environments. Infrared detectors are designed to operate at low temperatures between 2 and 170 K to prevent thermal excitation. This is important as thermal excitation can result in noise interference, reduced detector performance, and poor imaging quality of the object being targeted [37]. The applications of emissivity / absorptivity are illustrated in Fig. 2.

Measuring the thermal radiative properties of materials at cryogenic temperatures poses a challenge due to the weak radiation signals, primarily above 20 μm [38], emitted by the materials and the additional uncertainty introduced by background radiation. Addressing this challenge requires a concerted effort to develop more accurate measurement techniques and methods while minimizing ambient radiation.

2. Methods

To determine emissivity or absorptivity at cryogenic temperatures,



Fig. 2. Applications of emissivity/absorptivity measurement at cryogenic temperatures. (a) Medical application, (b) Cryogenic research, (c) Instrument design, (d) Space exploration, (e) Planetary exploration and remote sensing, (f) Cryogenic storage and transportation.

two main measurement techniques are used. The first is the calorimetric method, which calculates the total hemispherical emissivity of a sample across all wavelengths and angles by measuring its heat loss or gain with a calorimeter. The second is the radiometric technique, which compares the sample's reflected or emitted radiation with blackbody radiation at the same temperature using a bolometer. This method can measure emissivity at various angles (directional emissivity/absorptivity) and wavelengths (spectral emissivity/absorptivity). The measurement techniques are illustrated in Fig. 3.

2.1. Calorimetric techniques

The principle of calorimetry is based on the law of energy conservation, can be stated as the amount of energy a sample absorbs must equal the amount released. Calorimetric methods measure the heat lost or gained by the sample and can only estimate the total hemispherical values of emissivity or absorptivity.

The calorimetric technique can be sub divided as the steady-state method and the transient method. The steady-state method was predominantly employed for applications involving high temperatures exceeding 700 K; the surface is electrically heated to the desired temperature (high temperature) while it radiates heat to a liquid nitrogen-cooled black body [39–42]. However, nowadays, the steady-state method for cryogenic temperature emissivity measurements is also used [43–59]. The transient method involves heating the setup to a temperature higher than the desired one, removing the heat source. This process allows the sample to cool down and radiate heat. The sample change in temperature, mass, and heat capacity is used to calculate the emissivity of the sample [60]. Table 1 shows the characteristics of emissivity/ absorptivity measurement devices using calorimetric techniques.

Using a calorimetric technique, Giulietti et al. [61] measured the emissivity and absorptivity of high-purity metals and coatings at low temperatures, including Copper, Aluminum, and Tantalum. Measuring emissivity directly at low temperatures is challenging, so the thermal property of solids (expansion) was utilized instead of directly measuring

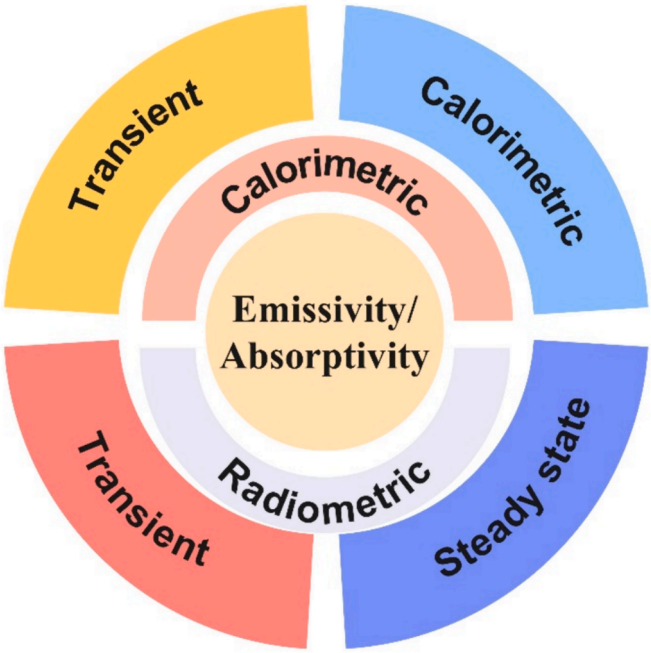


Fig. 3. Measuring methods for emissivity.

Table 1
Characteristics of emissivity/ absorptivity measurement devices using calorimetric techniques.

	Author (year)	Temperature/ K	Material	Thermal radiative property	Ref.
1	Giulietti et al. (1981)	77–343	Al, Tantalum, Cu	Emissivity and absorptivity	[61]
2	Amano et al. (1987)	High (70–85) Low 4.3	Al. polished with (SS*, SS* pol, SS* + Cu), SS* + Al (taping), Al pol + Al mylar, Cu oxidized, Cu polished)	Emissivity	[62]
3	Cunnington et al. (1970)	60–300	Gold on mylar, gold on Kapton, gold on Cu	Emissivity	[43]
4	Anderson et al. (1971)	0.9–6	304 SS*, Cu, gold + SS*, epoxy powder composite	Emissivity	[44]
5	Jaworske (1993)	173–673	Al + Zinc oxide	Emissivity	[45]
6	Brumley et al. (1996)	4.2–11	Nb film	Emissivity	[63]
7	Christophe Fabron (2000)	85–300	PCBZ white paint and Z306 black paint	Emissivity	[46]
8	Kralik et al. (2004)	High (10–120)Low (5–10)	Al, Cu	Emissivity and absorptivity	[47]
9	Duckworth et al. (2006)	25–35	Cu + silver (water contamination)	Emissivity	[48]
10	Hameury et al. (2007)	253–473	304 SS*	Emissivity	[49]
11	Kauder, Lonny R. (2009)	50–340	Al + Aeroglaze Z306	Emissivity	[50]
12	Tuttle et al. (2012)	20–300	Ball infrared black™	Emissivity	[51]
13	Yenni et al. (2012)	70–200	Aeroglaze, PUC**	Emissivity	[52]
14	Kim et al. (2014)	80	Al, Cu	Emissivity	[53]
15	Tuttle et al. (2014)	20–300	Aeroglaze Z307, Black Kapton, Ball Infrared Black™	Emissivity	[54]
16	Kralik et al. (2016)	20– 300	Pure Al	Emissivity and absorptivity	[55]
17	Hatti et al. (2018)	90–250	SS*, Cu, Kapton	Emissivity	[56]
18	Yenni et al. (2019)	82–150	Aeroglaze Z307	Emissivity	[57]
19	Shen et al. (2019)	15.4–300	–	Emissivity	[58]
20	Ando et al. (2019)	20–80	Black paint, carbon nano tubes	Emissivity	[64]
21	Zhang et al. (2022)	50–300	Carbon coating	Emissivity	[59]

* SS (Stainless- steel).
** PUC (Polyurethane coating).

temperature using pure metals. By comparing experimental findings with theoretical predictions, this study highlights the discrepancies and potential causes, such as low-intensity radiation detection challenges and surface conditions. This was a positive step towards developing more accurate and reliable methods for emissivity measurement.

Using the boiloff method Amano et al. [62], the emissivity of metal and its coating at low temperatures was experimentally measured. Fig. 4 presents the equipment used for experimental investigation.

The cold plate uses multilayer insulation, and heat transfer from the warm box to the cold plate is measured using Eq. (7)

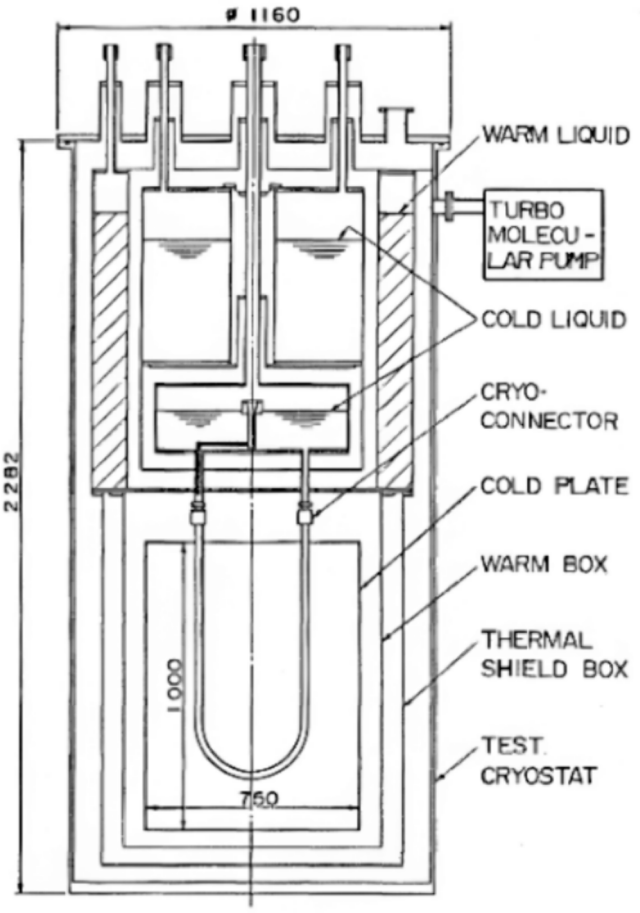


Fig. 4. Scheme of the experimental system [62].

$$Q_R = m\frac{\rho_l}{\rho_l - \rho_g}h_v - Q_{BG} \tag{7}$$

Here Q_{BG} is the heat loss, m is the flow rate of the gas evaporated, ρ_l and ρ_l are densities of liquid and gas in saturated conditions, respectively.

The comparison of experimental values and numerical calculations across various layers of multilayer insulation demonstrates a high level of agreement, as shown in Fig. 5.

Cunnington et al. [43] used a calorimetric apparatus to measure the emissivity of thin metallic films at cryogenic temperatures ranging from 60 K to 300 K. The setup involved mounting substrates in a vacuum chamber cooled with liquid nitrogen or hydrogen, with temperature precisely controlled by a spiral heater and a platinum resistance thermometer. The emitted energy from the sample was captured by a blackbody absorber, as shown in Fig. 6.

Using a calorimetric method, Anderson et al. [44] investigated the measurement and control of thermal radiation below 6 K, providing critical insights into the emissivity of various materials and the effectiveness of thermal shields in cryogenic environments. The power absorbed by a bolometer at 0.2 K was measured when illuminated by a source at 0.94 K, employing an outer cylindrical copper can as the radiation source and a flat rectangular vane bolometer for detection, as shown in Fig. 7.

Various materials were tested as source liners, including type-304 stainless steel, copper, gold-plated stainless steel, and an epoxy-copper powder composite. Stainless steel showed an emissivity of about 0.07, while copper and gold-plated stainless steel had temperature-dependent emissivities. Several thermal shields were also evaluated, such as Mylar, silvered Mylar, and coil-foil, with silvered Mylar significantly reducing

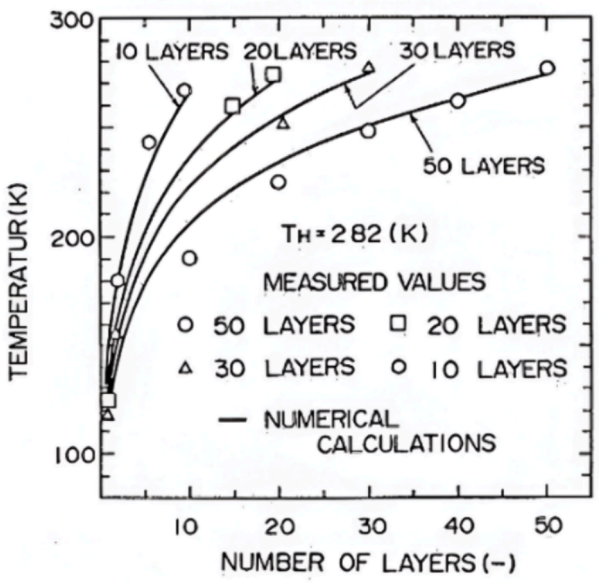


Fig. 5. Temperature distribution of multilayer insulation system [62].

thermal radiation. The epoxy-copper powder composite proved most effective, cutting absorbed power by 50 %.

Jaworski [45] introduced an emissivity measurement technique based on a finite element analysis model using both transient and steady-state calorimetric techniques. This method measures thermal emissivity for a 173–673 K temperature range. The equipment used for the experiment includes a round-shaped specimen suspended by a slim thermocouple wired within a testing compartment under vacuum conditions. The specimen emits radiation towards a cold wall cooled by liquid nitrogen. The sample was made of aluminum and coated with zinc oxide, which ensured that the emissivity values of more than 0.5 were accurate within a probable error of less than 4 %. The aluminum sample was illuminated using radiant energy generated by a quartz halogen lamp, as shown in Fig. 8.

The thermal emissivity between a superconducting niobium film (Nb) at 4.2 K and a titanium (Ti) film at temperatures ranging from 4.2 K to 11 K was experimentally measured by Brumley et al. [63]. The effective emissivity of the Nb film was evaluated at approximately 3 %. An Nb-coated quartz sphere was utilized as a bolometer to measure the total heat absorbed, and effective emissivity was derived from these measurements. The study findings are relevant to cryogenic applications, especially in space missions involving superconducting materials. The findings indicate that the interaction between Nb and air impacts its superconducting properties and emissivity.

Using a calorimetric approach, Fabron, Christophe [46] investigated an advanced method for measuring low hemispheric emissivity at cryogenic temperatures. The emissivity was determined by heating a suspended sample in a vacuum chamber and measuring the radiative power exchange in a controlled environment, as shown in Fig. 9.

The results for PCBZ white paint and Z306 black paint demonstrate the test bench's efficacy, validated by comparisons with data from other reputable laboratories. This work is crucial for improving thermal management in satellite components, providing essential emissivity data that underpin developing and operating advanced space observation instruments like the "Infrared Atmospheric Sounding Interferometer (IASI)".

Kralik et al. [47] presented a device as shown in Fig. 10, using two parallel surfaces and calculated emissivity by substituting radiative heat flow between them using a heater. The working range of the device was between 10 and 120 K of the temperature of the radiating sample, and the temperature range of the absorbing sample was 5 – 10 K. This method provides a precise way to evaluate the mutual emissivity of metals at low temperatures. The equation used to calculate the mutual emissivity of the two plates is given as follows,

$$\varepsilon_{RA} = \frac{Q_R}{A\sigma(T_R^4 - T_A^4)} \tag{8}$$

Here, T_R and T_A denotes the temperature of the radiator and absorber, respectively, A is the area of the absorber (same as the radiator), and σ is the Stefan-Boltzmann constant.

Duckworth et al. [48] designed and fabricated an experimental arrangement to accurately determine the emissivity of uncontaminated and contaminated metallic surfaces within a specific temperature range of 25 – 35 K. A copper sample plated with silver was utilized, with a surface area of 0.25 m^2. Radiation heat transfer between the thermal shield and the testing material was measured to calculate the emissivity of the sample, which is placed on the head of a cryocooler through a heat meter and enclosed by an aluminum shield covered by Vel-Black®. The aluminum shield acts like a blackbody radiation source with an emissivity of 0.95 [65]. The heat load of the sample can be measured with a heat meter developed by Kuchnir [66], as shown in Fig. 11.

The emissivity of the sample was calculated using a statement for the radiation heat transfer between two bounded surfaces.

$$\varepsilon_{sample} = \left[\frac{\sigma A_{sample}\left(T_{shield}^4 - T_{sample}^4\right)}{Q_{sample}} - \frac{A_{sample}}{A_{shield}}\left(\frac{1-\varepsilon_{shield}}{\varepsilon_{shield}}\right)\right]^{-1} \tag{9}$$

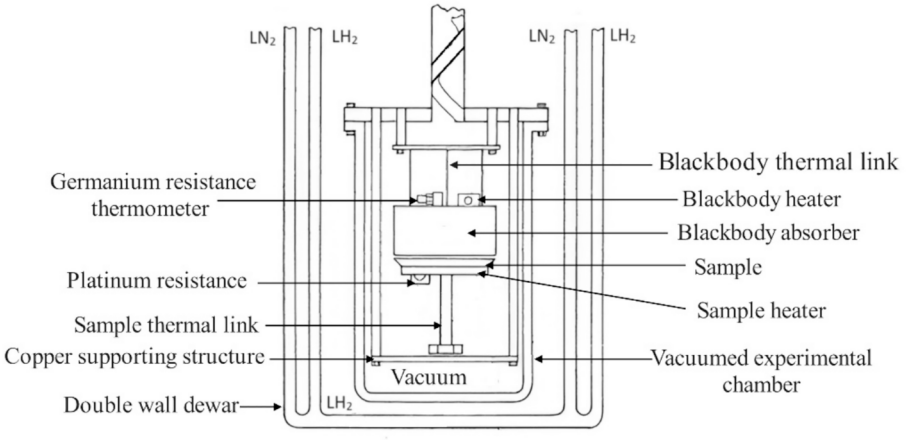


Fig. 6. Scheme of the experimental setup [43].

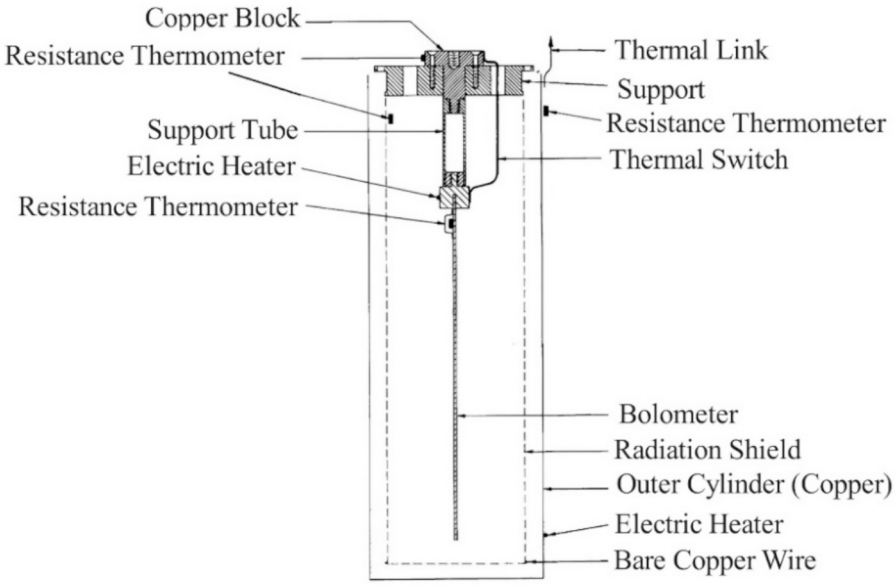


Fig. 7. Schematic of the system [44].

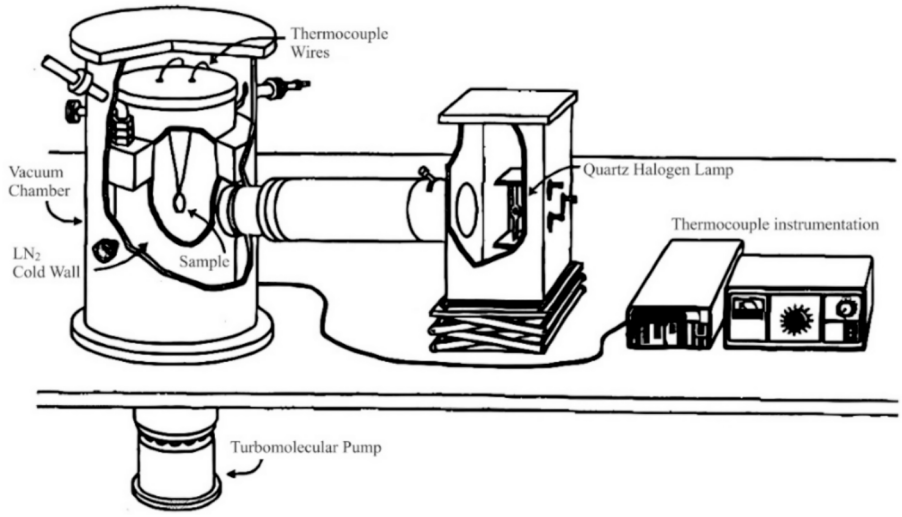


Fig. 8. Schematic diagram of low-temperature calorimetric vacuum emissometer [45].

Here, where σ is the Stefan-Boltzmann constant, ε_{shield} is the effective emissivity of the black thermal shield, ε_{sample} is the material's emissivity, A_{Shieid} is the area of the shield, and A_{sample} is the area of the sample.

Copper samples that were silver-plated were cooled down, and their emissivities were measured at 30 – 31 K. These values were then compared to room temperature measurements, which were made with a radiant-based emissometer, and to emissivities previously measured at 77 K (by Flynn [67]) and 300 K (by Ageladarakis [68]) for similar materials.

Hameury et al. [49] introduced a novel calorimetric technique for measuring the total hemispherical emissivity of opaque solid materials within a temperature range from −20 to 200 °C. The apparatus employs two identical samples and thermal guard rings to ensure one-dimensional heat flow, significantly reducing heat-loss corrections. The process involves extrapolating surface temperatures from two points on each sample, using a model that accounts for the main radiating surfaces in the chamber. The scheme of the system is presented in Fig. 12.

The apparatus achieves an expanded uncertainty range of ±0.005 to ±0.03, validated through comparison with other established methods, offering high accuracy, especially for materials with high emissivity and low thermal conductivity.

The novel emissivity measurement method introduced by Kauder [50] has been shown to enhance the accuracy and reliability of assessing the thermal properties of spacecraft coatings, which is crucial for optimizing spacecraft thermal management. Traditional calorimetric techniques used for determining the emittance of thermal control coatings have a significant flaw, which Kauder's work has addressed. By revising the heat balance equation, emittance is accurately measured by this methodology, which is especially beneficial when the sample and shroud wall temperatures are similar. Experiments at NASA Goddard Space Flight Center, utilizing an Aeroglaze-coated aluminum substrate within a cryogenic shroud, have validated this improved theoretical formulation, offering a promising avenue for enhancing spacecraft performance through better thermal control systems.

A simple yet relatively inexpensive approach by Tuttle et al. [51] for measuring the emissivity of BIRB is crucial for designing efficient thermal control systems in space-flight applications. The total hemispherical

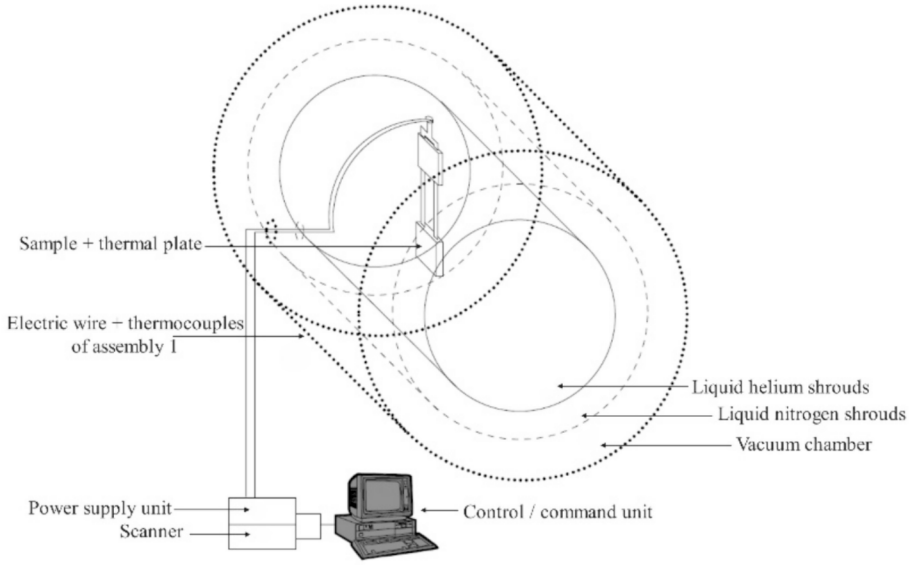


Fig. 9. Schematics of the system [46].

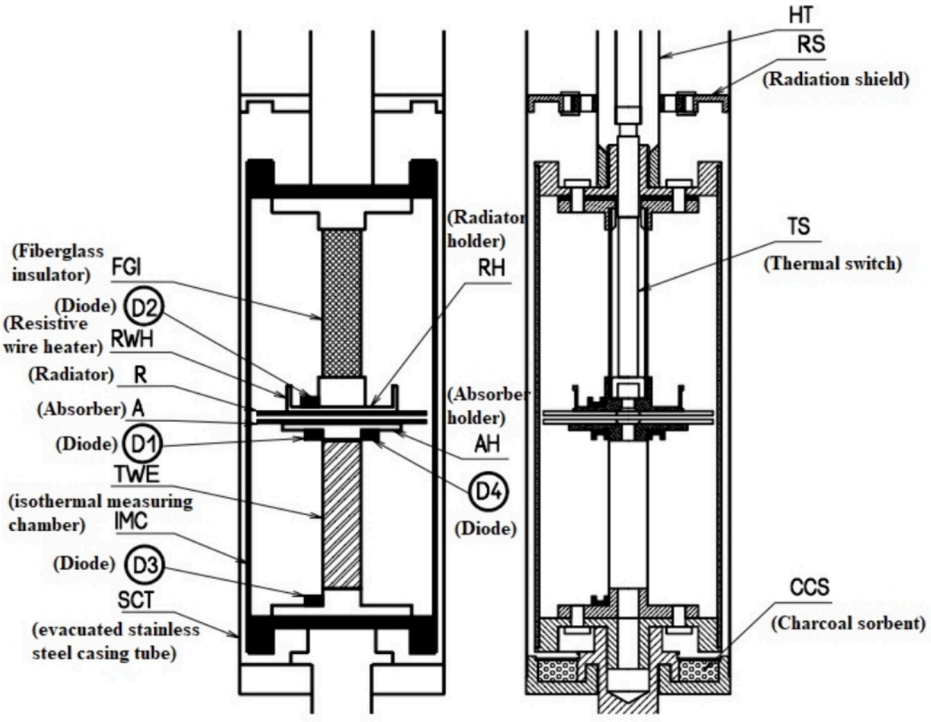


Fig. 10. Shows the lower part of the system [47].

emissivity of black surfaces is about 1 above 100 K but decreases steeply below 100 K. The spacecraft instruments must be passively cooled on space missions for regular operation below 40 K.

For precise measurements, both the absorber and emitter were coated with the same material, "BIRB (Ball InfraRed Black TM)", produced by "Ball Aerospace and Technologies Corp (BATC)" [69]. Fig. 13 represents the schematic of the system.

The results showed a weak temperature dependence with no steep drop-off at low temperatures, indicating higher emissivity values than previously measured [70] at "NASA's Goddard Space Flight Center" using large flat-coated samples hanging in a helium-cooled shroud inside a large vacuum chamber as shown in Fig. 14.

Using a steady-state calorimetric method, Yenni et al. [52] investigated the total hemispherical emittance of spacecraft thermal control coatings at low temperatures, specifically Aeroglaze and PUC. The emissivity values for Aeroglaze in the temperature range of 80 – 150 K and for PUC from 70 K to 200 K were accurately measured, with uncertainties reported to be less than 4.5 % and 7.4 %, respectively.

Kim et al. [53] presented an experimental method for measuring the emissivity of coated copper and aluminum samples at 80 K using a heat balance technique where the sample is suspended in a vacuum chamber and heated by a known power. The experimental system is shown in Fig. 15.

The results show that lampblack varnished coatings have high

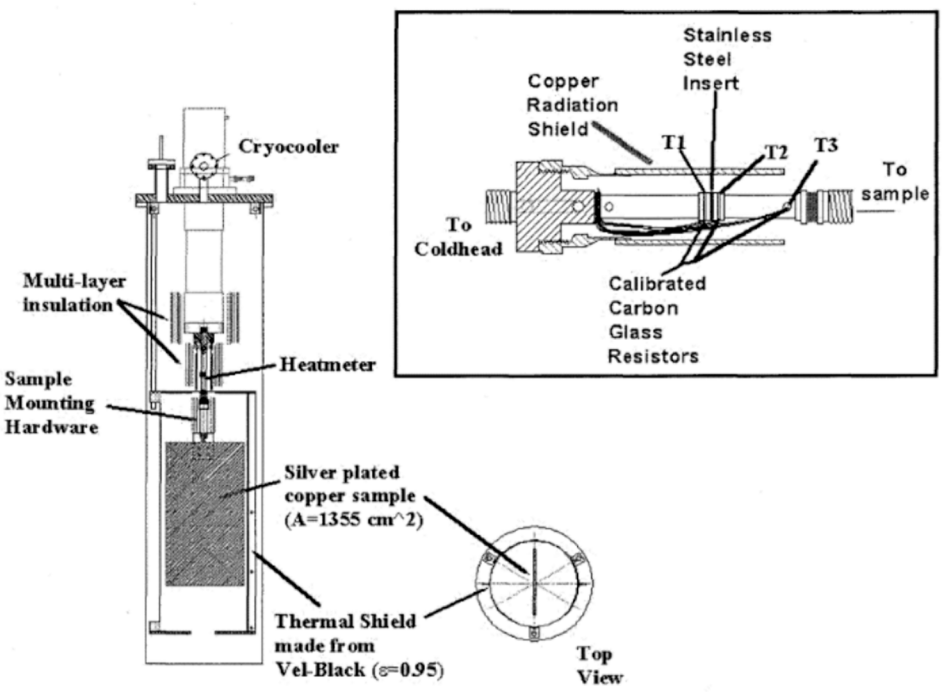


Fig. 11. Diagram of emissivity experimental test facility with inset of stainless steel/copper heat meter (top, right) and the geometry of the silver-plated copper sample relative to thermal shield from an overhead view (bottom, center) [48].

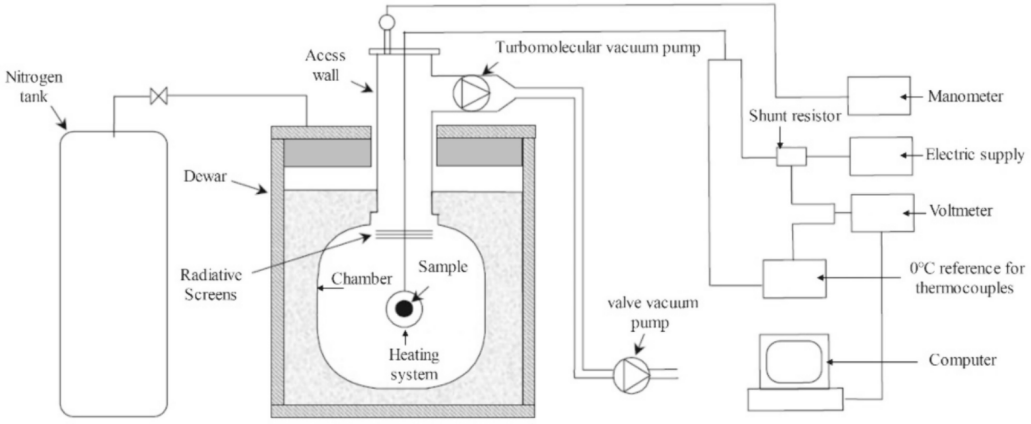


Fig. 12. Scheme of the system at 0 ℃ [49].

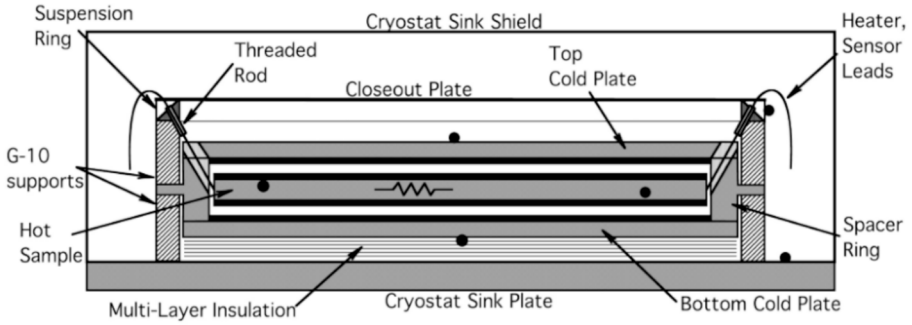


Fig. 13. The Schematic of the system. Thick black lines depict the graphical representation of the BIRB, while the thermometers are represented by black dots [51].

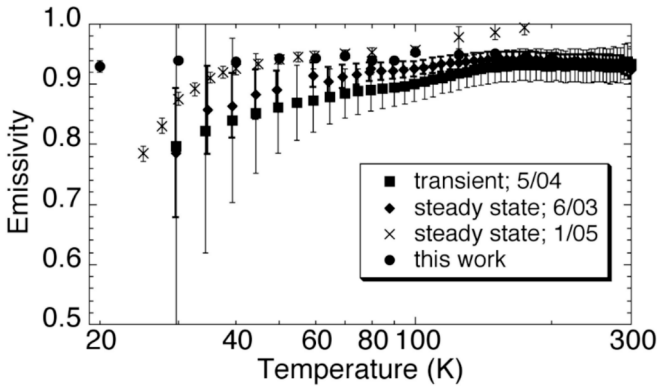


Fig. 14. Comparison of current data results with previous measurements, all taken at NASA's Goddard Space Flight Center [51].

emissivities (0.73 – 0.79), while bare copper and aluminum plates have lower emissivities (0.06 – 0.20), with the error analysis, including the contributions from radiated heat leak, conductive and convective heat leaks, instrument errors, and statistical errors.

To measure total hemispheric emissivity at 30 K, Tuttle et al. [54] from NASA modified an old apparatus [51] by including a honeycomb on both cold plates, as presented in Fig. 16. NASA missions used a nearly perfect radiator ($\epsilon \approx 1$) to emit heat into deep space to cool the instruments down to their cryogenic temperatures.

A high-emissive coating called Chemglaze™ Z307 [71] was applied to the honeycomb cells commonly used in spaceflight applications. The honeycomb was painted due to its convoluted geometry and high total hemispheric emissivity. Table 2 presents the three different characterized honeycomb configurations.

The differences observed among the three honeycomb models are minor and fall well within the range of error bars. Furthermore, all three honeycomb models exhibit emissivities above 0.95 at extremely low temperatures of 30 K. This technique offers a high-precision and non-intrusive method for measuring the emissivity of complex geometries

Table 2
The details of the three examined honeycomb specimens [54].

Specimens #	Core thickness (mm)	Cell size (mm)	Core Foil Thickness (μm)	Avg. Core coating thickness (μm)
1	12.7	3.175	38.1	16.3
2	9.53	3.175	50.8	17.0
3	9.53	3.175	17.8	8.4

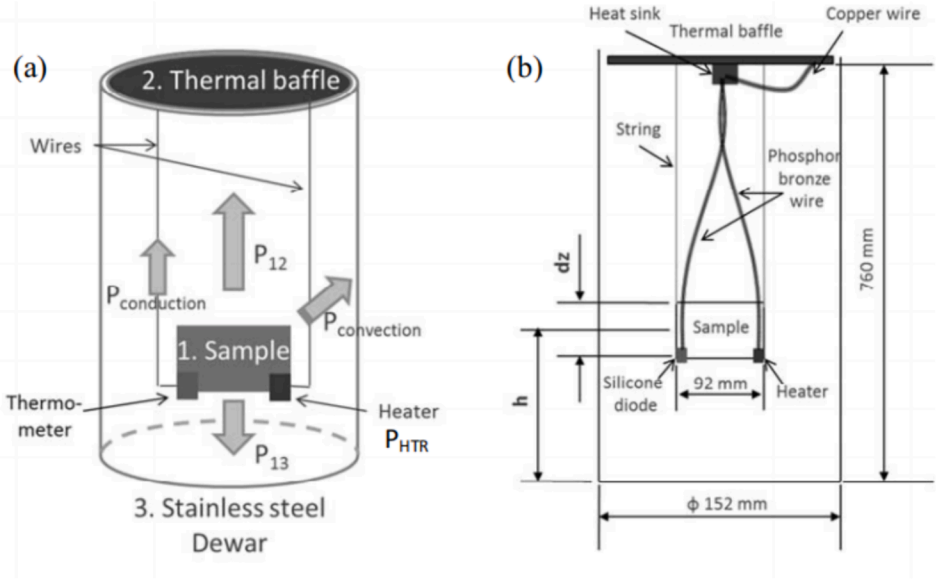


Fig. 15. The Scheme of the system. (a) Display thermal loads (b) displays the dimensions used to determine the view factor [53].

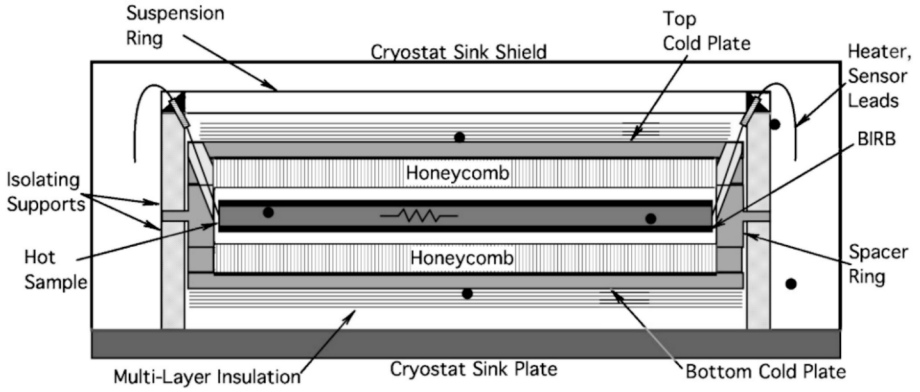


Fig. 16. A schematic view of the modified testing system. Thick black lines depict the graphical representation of the BIRB on a hot sample hanging between a honeycomb structure, while the thermometers are represented by black dots [54].

like honeycomb. It has important implications for the design of cryogenic systems such as the James Webb Space Telescope (JWST) [72].

Kralik et al. [55] modified the apparatus presented in 2004 [36] to determine total hemispherical emissivity at cryogenic temperatures (down to 20 K), measuring the emissivity and absorptivity of highly reflective surfaces. The instrument description is given in Fig. 17.

The method's core involves using a heat flow meter to gauge radiative heat transfer between two plane parallel disc samples placed in a vacuum environment. The tested sample is maintained at cryogenic temperatures, while a reference surface with known properties is a comparator. The "Emister" instrument employs a thermal resistor linked to a liquid helium (LHe) bath for heat flow calibration. The method's reliability is underscored by comprehensive uncertainty analysis, demonstrating an uncertainty below 11 % for emissivity measurements at around 30 K for pure aluminum, improving to below 7 % for measurements above 60 K.

Hatti et al. [56] presented a novel system that utilizes a calorimetric technique to measure the total hemispherical emissivity of metallic and non-metallic tapes and paints at low temperatures, up to 90 K. The setup consists of a cryostat, a vacuum chamber, a turbo-molecular pump, thermal baffles, and a heater. The cryostat is a cylindrical vessel that holds liquid nitrogen, reducing radiation and gas conduction through super insulating material and an evacuated intermediate space. The vacuum chamber, positioned inside the cryostat and cooled by liquid nitrogen, provides the test area for emissivity measurements. The system uses a stainless-steel heater to bring samples to the desired temperature and measure emissivity through heat flux calculations. The results show the emissivity values for stainless steel, copper, aluminum, Kapton tape, aluminized Mylar, and black paint, aligning closely with literature values and demonstrating the setup's accuracy and reliability.

Yenni et al. [57] developed a calorimetric experimental approach to iteratively estimate the emissivity of a cryogenic shroud and unknown test samples at very low temperatures. This approach uses a steady-state energy balance involving a plane surface and a cavity surface substrate coated with the test sample. This approach has been validated by estimating the emissivity of a cryogenic shroud coated with ECP2200 paint and a test sample coated with Aeroglaze Z307 paint, which demonstrates its potential for measuring the emissivity of unknown test samples at very low temperatures.

Shen et al. [58] developed an innovative cryogen-free equipment to find the material's emissivity rapidly at low temperatures (between 15.4 K and room temperature), consisting of a two-stage GM cryocooler and a dismountable sample test rod independent of the vacuum environment, as shown in Fig. 18.

Without disturbing the vacuum, the sample can be changed, making it a rapid emissivity measurement device and an attractive solution for measuring the thermal emissivity of materials. The emissivity of material is calculated using the steady-state calorimetric method and the following relation.

$$Q_R = A\sigma(\epsilon_R T_R^4 - \epsilon_A T_A^4) \tag{9}$$

The equation describes the radiative heat flow between a radiator and an absorber, where σ represents the Stefan-Boltzmann constant. The variables ε_R, ε_A, T_R, and T_A represent the emissivity and temperature of the radiator and absorber, respectively. Q_R denotes the radiative heat flow between the two bodies, and A represents the radiative area of the radiator and absorber.

Ando et al. [64] used the calorimetric method to investigate the emissivity of various black coatings and an open honeycomb structure at cryogenic temperatures ranging from 20 K to 80 K. The investigation involved a vacuum vessel with aluminum alloy shields that were maintained at low temperatures, which allowed precise measurements of total hemispherical emissivity to be made. The method's principle is based on the quantification of the thermal radiation that was emitted by the test samples, which included black paint, carbon nanotubes (CNT), and black plating, under controlled conditions. It was found that the emissivity of the black-plated open honeycomb structure could reach values as high as 0.9, significantly outperforming the uncoated surfaces, which exhibited an emissivity of around 0.5. This study serves to highlight the critical role played by material selection and structural design in the enhancement of thermal performance, offering valuable insights for applications within cryogenic systems and aerospace technology.

Zhang et al. [59] presented a method for measuring total hemispherical emissivity from 50 K to 300 K based on the static calorimetric method. The principle of this apparatus is similar to Kralik et al. [47], using two parallel surfaces with a small distance between them. However, a two-stage GM cryocooler was used instead of a liquid helium bath.

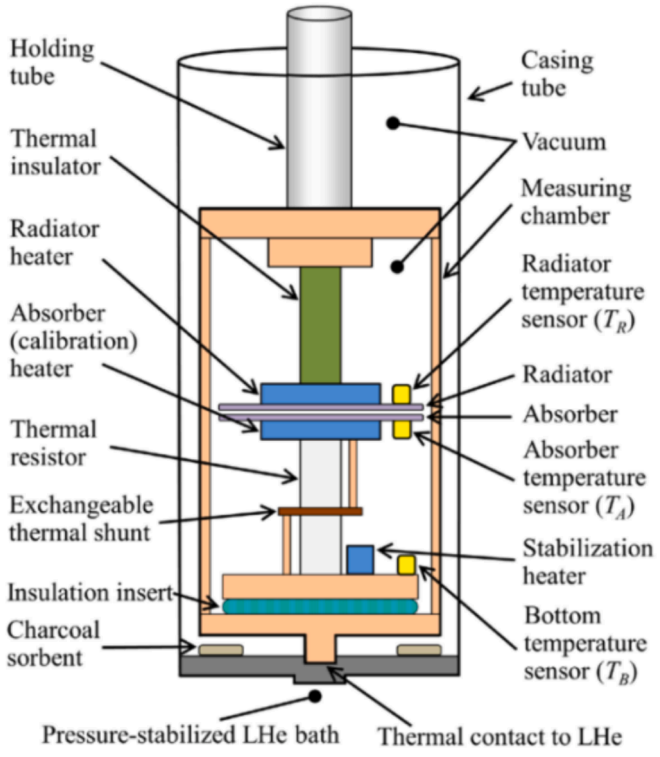


Fig. 17. The Schematic of the modified testing system [55].

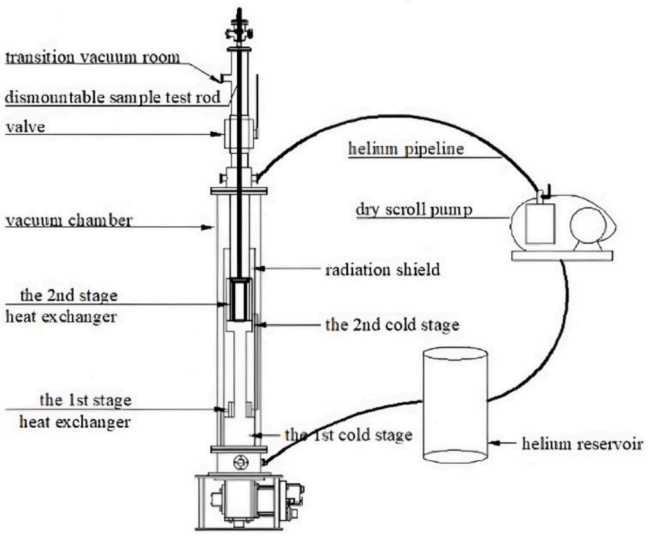


Fig. 18. The scheme of a cryogen-free system [58].

The apparatus consists of a two-stage GM cryocooler, testing room, thermal switch, heat-conducting rod, vacuum shield, radiator shield and flange, as shown in Fig. 19.

This method measured the thermal emissivity of a carbon coating, which showed a temperature dependence of 0.7 at room temperature and 0.52 at 50 K, offering a promising approach for characterizing the thermal radiative properties of materials at cryogenic temperatures.

In summary, Giulietti et al. [61] employed a unique method to measure the temperature of metals through thermal expansion. Similarly, Shen et al. [58] and Kralik et al. [55] utilized a comparable approach to assess the emissivity and absorptivity of materials using two parallel plates placed at a small distance. Tuttle et al [54], Yenni et al [57], and Ando et al. [64] investigated the emissivity of black paint and experimented with different honeycomb structures for potential use in space satellites. Kauder et al. [50] and Kim et al. [53] suspended samples using strings or wires in the calorimeter to measure emissivity; however, this technique requires accounting for the view factor, adding complexity to the measurements. Overall, these studies highlight the diversity of approaches in accurately measuring emissivity and absorptivity, each with its specific challenges and applications.

2.2. Radiometric techniques

The radiometric technique detects reflected or emissive radiation using a sensitive bolometer or spectrophotometer; as the radiation emission is spread out across the electromagnetic spectrum, a bolometer detects the range that is needed according to the sample. Also, this method can detect the sample's radiation from different angles to get the desired results. The drawback of this technique is that the sample radiation is so small, mainly above 20 μm at cryogenic temperatures, and background radiation overcomes the sample signal, so it is difficult to minimize or remove the background radiation. Different methods can be used to reduce the background radiation, like placing the experimental setup in a nitrogen-cooled wall with a high vacuum and using a high absorbent material on the inner side of the apparatus wall, etc. The two methods to measure emissivity using the radiometric technique are the steady-state and transient methods.

To accurately determine an object's radiative properties at thermal equilibrium, the steady-state radiometric method involves initially heating the object to a precise and steady temperature. Following this, an infrared radiometer captures the radiative power the object emits. This data is then benchmarked against the emissions of a blackbody reference, which is kept at an identical temperature, to ascertain the object's emissivity via the ratio of its emitted power to that of the blackbody. This technique boasts high precision and offers a direct approach to measuring radiative characteristics. However, meticulous calibration of the radiometer and the constant maintenance of thermal equilibrium are necessary to ensure accuracy. The transient radiometric method involves subjecting an object to an abrupt temperature shift and continuously monitoring the resulting radiative energy emission with an infrared radiometer during the transition phase. The process allows for calculating the material's emissivity by analyzing the temporal radiative power data in conjunction with the object's known thermal properties. This technique offers quicker data acquisition than steady-state methods; however, it introduces complexity in data interpretation and an increased risk of inaccuracies stemming from the rapid temperature fluctuations [73]. Table 3 shows the Characteristics of emissivity measurement devices using the radiometric technique.

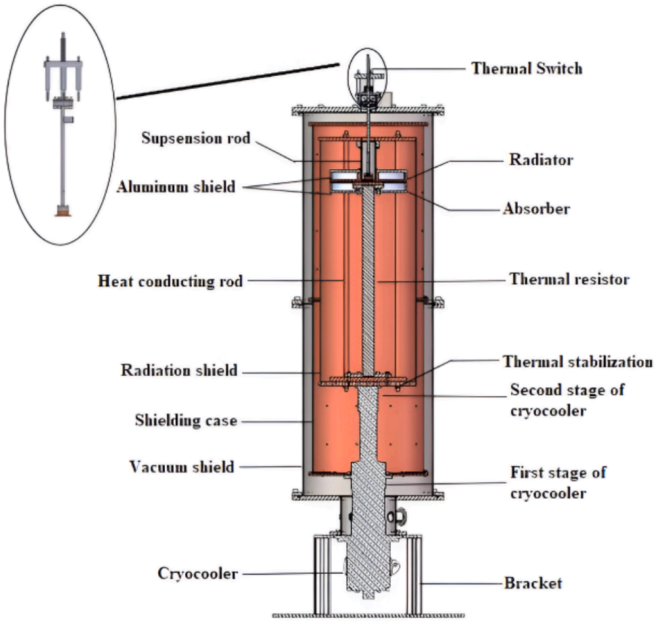


Fig. 19. Scheme of the system [59].

Spisz et al. [74] outlined an innovative method for concurrently measuring metals' total hemispherical emittance, solar absorptance, and the absorptance-emittance ratio at cryogenic temperatures. The approach compares how thin samples of metals like aluminum, copper, and 304 stainless steels react temperature-wise to a sinusoidal variation in applied radiant intensity. The experimental apparatus is set up in an environment with high vacuum and cold walls to keep gas conduction at bay, using a radiation source that could be either a 12-kW carbon arc or a 1-kW quartz-iodine tungsten filament lamp as shown in Fig. 20.

Bock et al. [75] developed a high-precision instrument to measure the emissivity of reflective surfaces at near-millimeter wavelengths. The instrument operates by comparing the thermal emission of a test sample to that of a calibrated reference surface, eliminating the need for direct heating of the sample. The system includes a superfluid helium composite bolometer detector with an ellipsoidal mirror setup, reflective chopper, and liquid-nitrogen-cooled load to modulate and measure thermal emissions, as shown in Fig. 21. The instrument boasts an absolute accuracy of $\Delta e = 5 \times 10^{-4}$ and can detect emissivity differences as

Table 3
Characteristics of emissivity measurement devices using radiometric technique.

	Author (year)	Temperature/ K	Material	Spectrum/ μm	Ref.
1	Spisz et al. (1968)	150–500	Al, Cu, 304 SS	0.38–2.5	[74]
2	Bock et al. (1995)	300	Al and gold coated on glass and carbon	1160–1330 (near millimeter)	[75]
3	Especel et al. (1996)	300	Black paint, duralumin, aluminum paint	1–50	[76]
4	Gentile et al. (1996)	5–300	–	0.2–20	[77]
5	Ishii and Ono (2003)	373	Black coatings (Astroblack, heat-resisting black, and Okitsumo B600) on metal substrates	5–12	[78]
6	Demiryont et al. (2006)	300	A thin layer of inorganic materials on glass and plastic substrates.	4–27	[79]
7	Herve et al. (2008)	20–200	Dielectric coatings are disposed of on a metallic substrate. (Velvet, Z306)	6–800	[80]
8	Woods et al. (2012)	80–300	Stainless Steel coated silver	0.4–200 (total)	[81]
9	Ma et al. (2019)	173–213	Stainless Steel, Quartz	8–14	[82]
10	Zhao et al. (2023)	240–300	Nextel 811–21, SS, G10	2–12	[83]

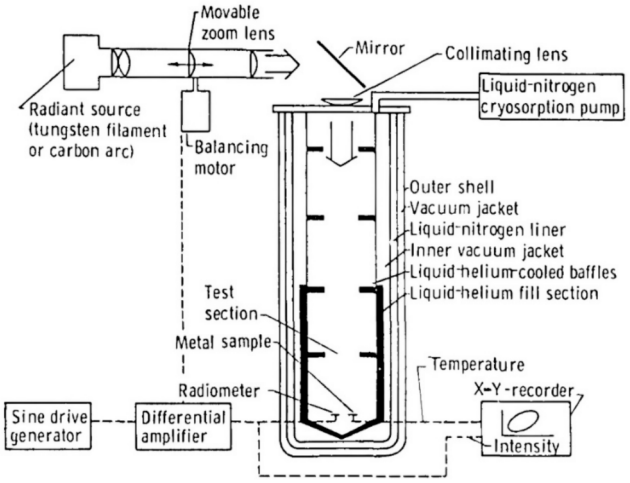


Fig. 20. Scheme of the experimental setup [74].

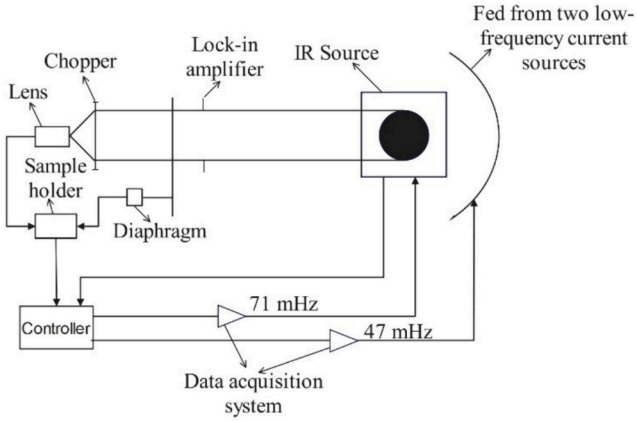


Fig. 22. Scheme of the experimental equipment [76].

slight as $\Delta e = 10^{-4}$. This capability is critical for applications in infrared and millimeter-wave telescopes, where reducing thermal emission from telescope components is essential.

The emissivity values obtained for various metal films, including gold and aluminum evaporated on glass and carbon fiber-reinforced plastic composites, demonstrated the instrument's precision. For instance, gold on glass exhibited an emissivity of 0.01215, while aluminum on CFRP showed 0.01265. This instrument's ability to measure emissivity accurately without heating makes it highly suitable for characterizing the thermal properties of advanced composite mirrors and other reflective materials used in space telescopes and other sensitive optical applications.

Innovative methods for measuring the total emissivity of materials at low temperatures without the need for an absolute reference were introduced by Especel et al. [76]. The focus was on implementing periodic radiometry techniques involving thermal modulation of both the sample and the infrared (IR) source to isolate emitted and reflected fluxes through direct and indirect methods using thermal modulation and Fourier transformation, as shown in Fig. 22. Experimental validation is conducted on samples with varying emissivity, demonstrating the reliability and accuracy of the measurements and consistency between the direct and indirect approaches.

Gentile et al. [77] developed the high-accuracy cryogenic radiometer (HACR) at the National Institute of Standards and Technology (NIST) represents a significant advancement in optical power measurement standards. Utilizing the principle of electrical substitution radiometry, the HACR compares temperature rises induced by optical and electrical heating within a highly absorptive copper cavity at cryogenic temperatures. The scheme of the radiometer is shown in Fig. 23. This design achieves an impressive relative standard uncertainty of 0.021 % at 0.8 mW, covering a wide spectral range from ultraviolet to infrared.

The study by Ishii and Ono [78] introduced a highly accurate Fourier-transform infrared spectrometer (FTS) developed by the National Metrology Institute of Japan for low-temperature thermometric applications. Utilizing a Michelson interferometer, the FTS covers a 5 – 12 μm wavelength range and employs advanced phase correction techniques for calibration, achieving a mean relative error of 0.6 %. The scheme of FTS is shown in Fig. 24. The FTS's ability to produce accurate Planckian spectra makes it a valuable tool in metrological and thermometric research.

For spacecraft thermal management applications, Demiryont et al.

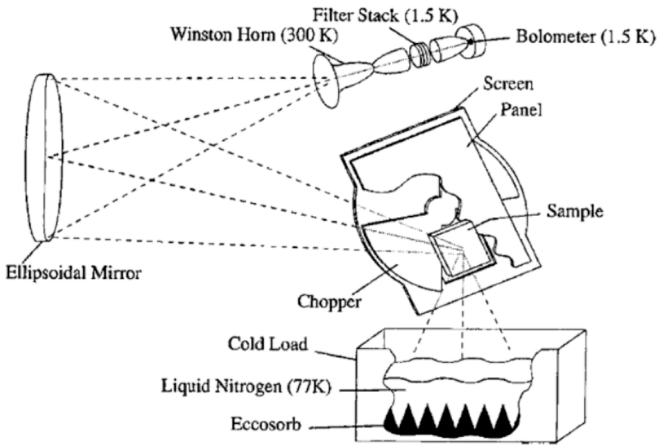


Fig. 21. Scheme of emissivitometer [75].

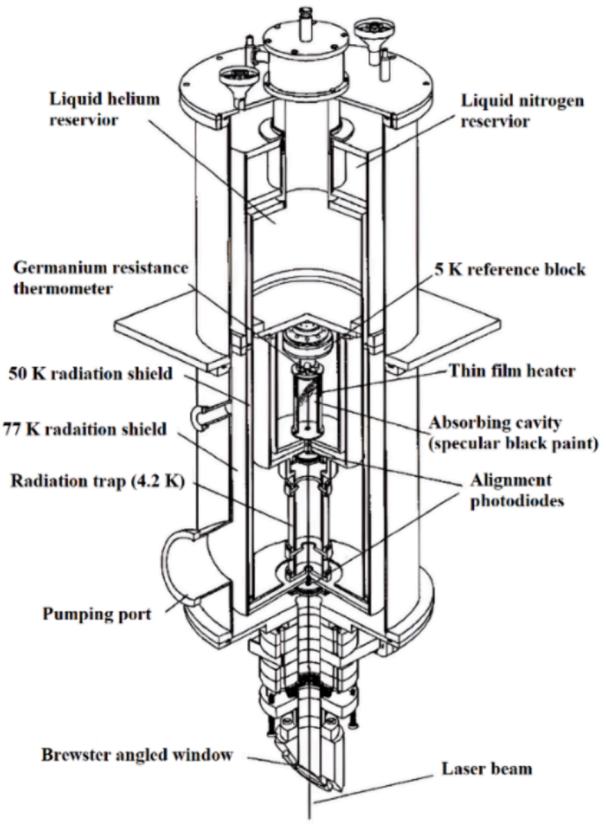


Fig. 23. Scheme of cryogenic radiometer [77].

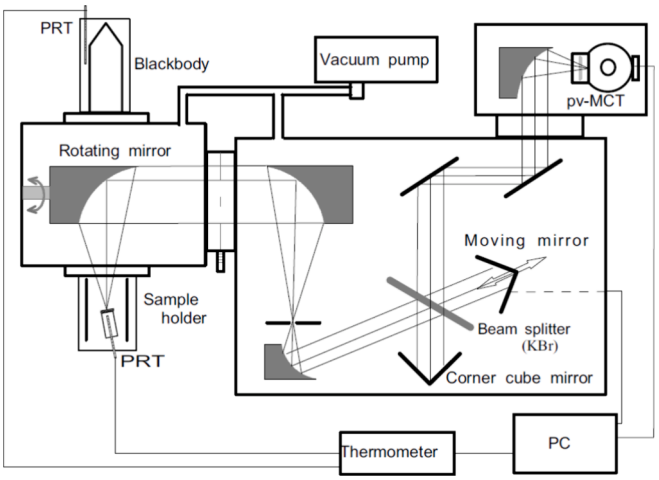


Fig. 24. Schematic of FTS [78].

[79] presented a novel Electrochromic Device (ECD) for satellite thermal control, offering a full range of emissivities and a solid-state, non-mechanical switch. The principle for emissivity measurement is approached by assessing the capability of the ECD to vary its emissive properties through voltage-induced changes, thereby allowing for the quantification of emissivity changes and the effectiveness of the device in controlling thermal radiation emission to be determined.

Using the iterative differential method, Herve et al. [80] presented a comprehensive method for directly measuring the total emissivities of satellite coatings at cryogenic temperatures, specifically between 20 K and 200 K, crucial for the thermal management of the Planck satellite. The setup features a lock-in amplifier with a chopper to modulate and isolate the signal from the sample, reducing interference from external radiation sources, as shown in Fig. 25.

Woods et al. [81] presented a method for measuring the broadband optical emissivity of highly reflective samples at cryogenic temperatures using a primary standard optical detector. The technique is significant for application in large-scale cryogenic systems, such as thermal shields in fusion reactors and satellite systems. The experimental setup involves a "National Institute of Standards and Technology (NIST)" "Absolute Cryogenic Radiometer (ACR)" with high absorptance from visible wavelengths to beyond 200 μm, ensuring low noise and high accuracy. A schematic of the entire measurement configuration is shown in Fig. 26.

The samples, including polished stainless-steel plates, are mounted in a helium-gas-cooled vacuum chamber, with background subtraction achieved via a cooled shutter and detailed temperature control.

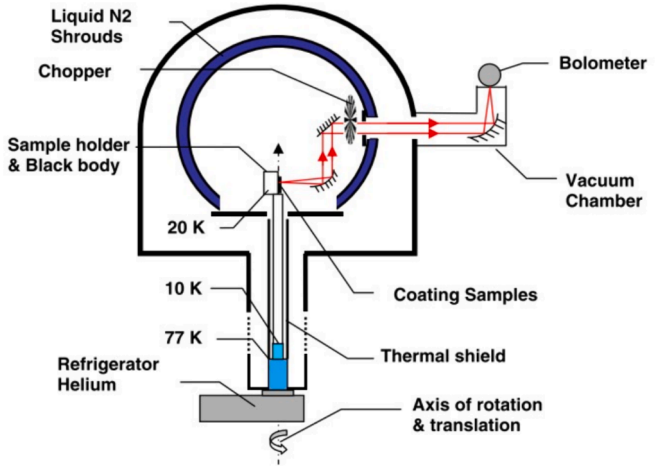


Fig. 25. Schematic diagram of the experimental setup [80].

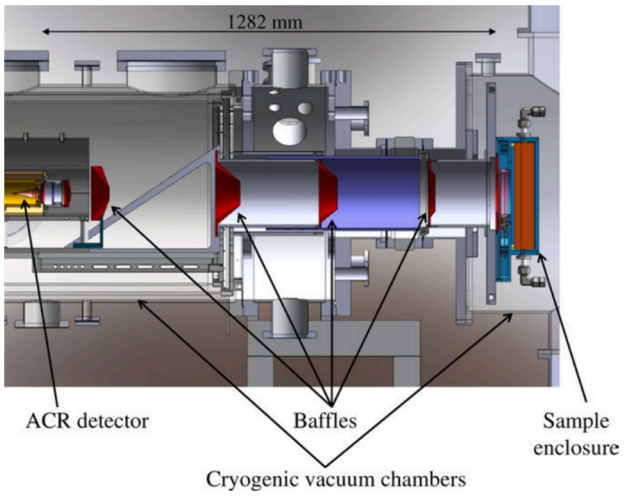


Fig. 26. Displays the configuration of the emissivity measuring setup [81].

Emissivity measurements are performed over the 80 – 300 K temperature range with an absolute uncertainty between 0.0002 and 0.002.

Ma et al. [82] designed an innovative apparatus to measure spectral emissivity at a wavelength of thermal control materials at low temperatures using FTIR (Fourier transform infrared spectrometer). The apparatus comprises an FTIR configured in step-scan mode, a sample cooling chamber, and a mechanical modulation system, including a chopper and lock-in amplifier, as shown in Fig. 27.

The primary advantage of this apparatus is its ability to distinguish between the spectral emissivity of different materials, providing high reproducibility and accuracy with a combined standard uncertainty of 3.3 % at 213 K.

Zhao et al. [83] preliminary designed a system presented in Fig. 28 for measuring emissivity at low temperatures using the direct radiometric method. By using GM (Gifford-McMahon) cryocooler as a cold source and MCT (mercury cadmium telluride) as a detector with an effective 2 – 12 μm bandwidth.

The blackbody, sample, and gold plane mirror were all maintained at the same temperature to determine the sample's emissivity. The system could switch between different radiation sources by employing a stepping motor, allowing for emissivity measurements of coatings such as Nextel 811 – 21, 304 stainless steels, and G10. To minimize ambient radiation effects, the system incorporated an off-axis parabolic gold mirror setup with an MCT detector and a cold optical path cover. The uncertainty analysis of the system indicated high accuracy, with an expanded standard uncertainty of less than 2.040 % at temperatures above 240 K.

In summary, the radiometric method is an effective approach for measuring the spectral emissivity and absorptivity of materials. Especel et al. [76] employed three different methods to determine emissivity, even achieving absolute emissivity measurements without the need for a reference. Researchers like Ishii and Ono [78] and Ma et al. [82] utilized

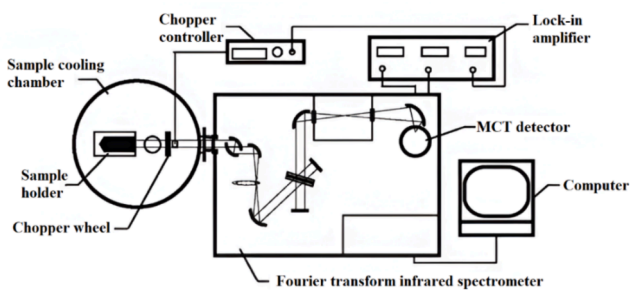


Fig. 27. Scheme of the emissivity measuring system [82].

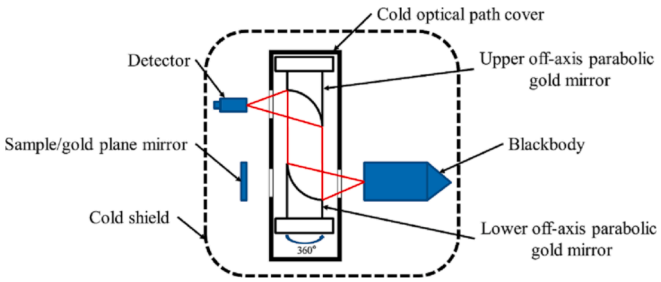


Fig. 28. Scheme of the system used to measure emissivity [83].

spectrometers for this purpose, with Ishii and Ono [78] achieving a relative mean error of 0.6 %, while Ma et al. [82] reported a combined standard uncertainty of 3.3 %. Other studies, including those by Bock et al. [75], Ma et al. [82], and Herve et al. [80], employed a nitrogen-cooled shield and chopper to enhance measurement accuracy. Additionally, Gentile et al. [77] and Woods et al. [81] demonstrated the use of high-precision cryogenic radiometers to achieve highly accurate emissivity measurements. Compared to the calorimetric method, the radiometric method not only requires less time but also achieves higher accuracy, particularly when using cryogenic radiometers.

3. Materials

Radiation heat transfer is crucial in low-vacuum conditions and cryogenic temperatures commonly encountered in space applications. It is vital to understand the thermal radiation properties of the materials used to design effective cryogenic devices. Aluminum and stainless steel are the most used materials for such devices. In this context, it is pertinent to discuss the emissivity/absorptivity of these materials in detail [84].

3.1. Aluminum

Aluminum is an important material used for the manufacturing of cryogenic devices. Here, aluminum's emissivity and absorptivity, which will be measured using different methods, are summarized.

Giulietti et al. [61] determined the emissivity of aluminum using the thermal expansion of metals and compared the results with theory and previously published data. Fig. 29 shows the two theoretical lines made by using the Hagen-Rubens relation [85] and Aschkinass (1905) relation [86]. The difference between theoretical and experimental increases is noticeable at lower temperatures.

Kralik et al. also used their experimental setup to find the mutual emissivity of aluminum and super-insulation. The mutual emissivity ε of aluminum foil shiny and matt face is measured compared to four foils provided by Austrian Aerospace (now Beyond Gravity), as shown in Fig. 30.

Using the same experimental setup as Kralik et al. [47] and Musilova et al. [90] measured the emissivity of aluminum samples (sheet, foil, mylar), aluminum alloys for the temperature range of 30–140 K. He further treated the surfaces using chemical, mechanical, and material annealing methods and analyzed their effect on thermal emissivity. The specifications of measured surfaces and the treatments are mentioned in Table 4.

The emissivity and absorptivity of aluminum are measured and compared the results with formerly published data in Fig. 31.

Fig. 32 compares various treatments on aluminum samples. Mechanical treatment reduced absorptivity by 30 % on 99.5 % pure aluminum, whereas chemical polishing achieved the lowest absorptivity value.

Fig. 33a shows that the absorptivity and emissivity of matt wrapping Al foil are slightly higher than those of the chemically polished Al sheets. Notably, the shiny side of the same Al foil exhibits 10–20 % higher absorptivity than the matt side. Furthermore, Fig. 33b illustrates the absorption and emissivity of aluminum samples with different surface treatments. The values of mechanically polished surfaces are like those of finely turned surfaces, and both are 20 – 50 % higher than the measured values of chemically polished surfaces. It is worth noting that emissivity values are higher than absorptivity at temperatures above 50 – 60 K, but both are equal at lower temperatures.

Kralik et al. [55] modified their system and presented the new results to validate and compare their present results on the emissivity of pure Aluminum samples with their measurement from 2005 [90] and with results obtained by other methods. Fig. 34 compares different Aluminum samples that were treated with various surface treatments, including chemical polishing (measured by current), electro-polishing [94], fine mechanical polishing with subsequent annealing [92], evaporated Al layers [91,95,96], and fine mechanical polishing. Despite the various methods used to measure emissivity, the experimental results generally exhibit high consistency.

Frolec et al. [14] used the instrument designed by Kralik et al. [55] to calculate the emissivities and absorptivity of metallic surfaces, mainly focused on aluminum. Different treatments were used on aluminum surfaces, and aluminum surfaces were covered with an ice layer. The emissivity strongly depends on the surface finish, as shown in Fig. 35.

The data on aluminum has been gathered through various methods and compiled into a summary. The collected information includes details on the emissivity and absorptivity of the material, which are

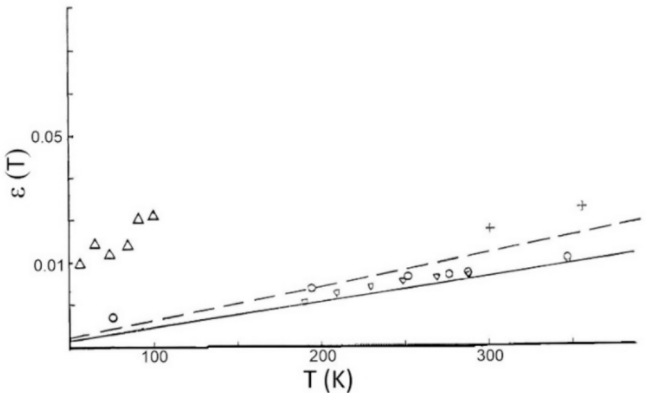


Fig. 29. Experimental and theoretical values of aluminum emissivity. ○ is current experimental data [61], Δ results belong to Hawks (1969)[87], + results belong to Jack and Spisz (1970)[88], ∇ belongs to Ramanathan et al. (1977) [89].

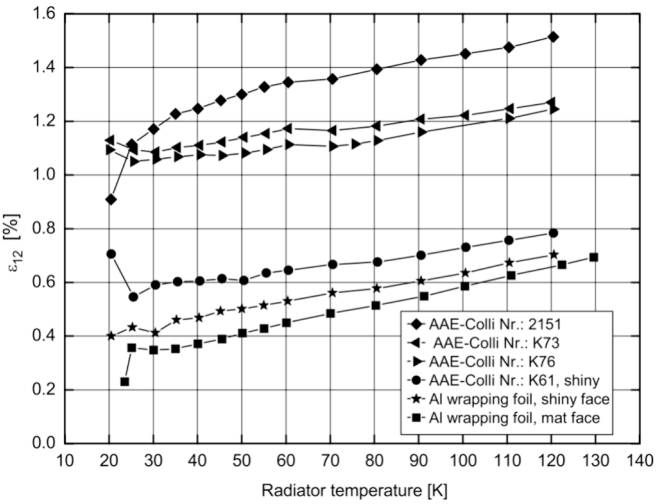


Fig. 30. The emissivity of some super-insulated aluminum foils [47].

Table 4
Specifications and treatments of measured surfaces [90].

No.	Surface[a]
Al 99.5 sheet	
04/2	Chemically polished, 1hr exposed to air
04/3	Sample 04/2 for half a year exposed to air in the laboratory
08/1	Unfinished
09/1	Vacuum annealed at 350°C for 20 min, then chemically polished, 20 min exposed to air
10/1	Abraded with stainless steel wool
11/1	Finely turned on a lathe
11/2	Sample 11/1 for a half year exposed to air in the laboratory
12/1	Chemically polished
12/2	Sample 12/1 for half a year exposed to air in the laboratory
Al foils 10 μ m thick	
07/01	Wrapping foil, shiny side
20/1	Foil (AAE-Colli No. K61[b]), shiny side
24/1	Wrapping foil, matt side
AAE (aluminized polyester foils[b])	
Nr. K73	6 μ m, DAM[c] CRYOGENIC standard
Nr. K76	12 μ m, DAM[c] CRYOGENIC standard
Nr. 2151	6 μ m, AM[c] CRYOGENIC standard, crinkled

[a] To enhance the quality, the samples were promptly placed in a vacuum within 24 h after surface treatment.
[b] Given by Austrian Aerospace (now Beyond Gravity).
[c] Double aluminized mylar.
[d] Aluminized mylar.

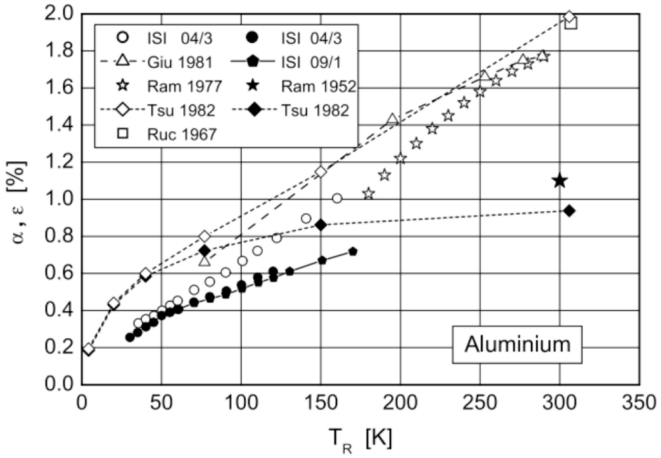


Fig. 31. Aluminum's total hemispherical absorptivity (represented by solid symbols) and emissivity (represented by open symbols). The data used to create the plot can be found in the table, and it was also sourced from other authors [61,89–93].

represented in Figs. 36 and 37, respectively. The figures showcase the behavior of aluminum under different temperature ranges, specifically between 20 and 350 K. The summarized data provides valuable insights into the behavior of aluminum, which can be beneficial in various fields, including cryogenics, material science, and engineering.

3.2. Stainless steel

It is worth noting that structural metals change properties when subjected to subzero temperatures from room temperature. These changes are most significant when the metal is cooled to extremely low temperatures close to liquid helium and hydrogen boiling points. However, certain types of stainless steels like 316 and 304 [97] are known to be "tough" at cryogenic temperatures, hence can be classified

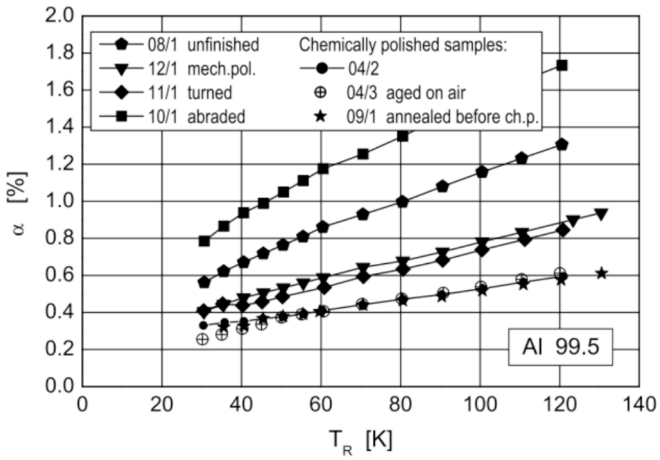


Fig. 32. Absorptivity of Aluminum sheets with several surface finishes [90].

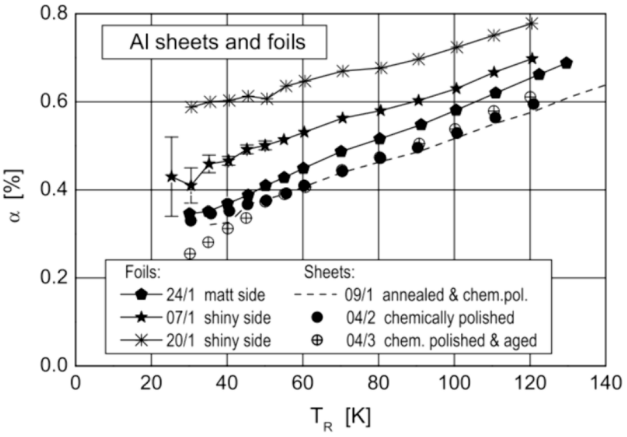


Fig. 33a. Absorptivity of Aluminum foils thickness 10 μ m and chemically polished Aluminum sheet purity (99.5) [90].

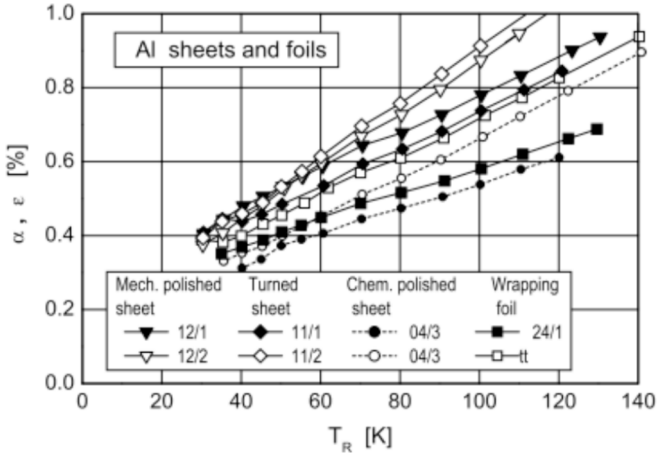


Fig. 33b. Absorptivity (represented by solid symbols) and emissivity (represented by open symbols) of Aluminum foil wrapping matt side compared with mechanically and chemically processed Aluminum sheets [90].

as "cryogenic steels" [98]; with this in mind, let's take a closer look at the emissivity of stainless steel when subjected to cryogenic temperatures.

Musilova et al. [55] used the same experimental setup as Kralik et al. [47] and found the emissivity and absorptivity of stainless steel and the

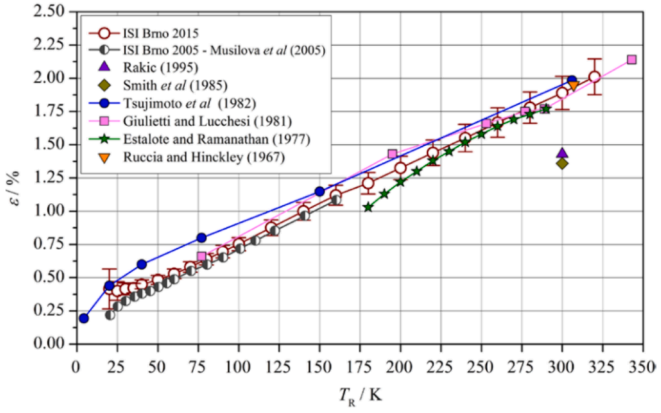


Fig. 34. Percentage emissivity of pure aluminum to radiator temperature [55].

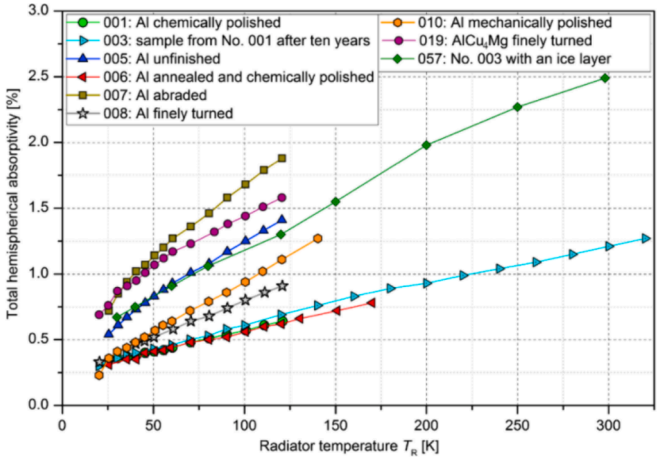


Fig. 35. Absorptivity of aluminum and duralumin samples [14].

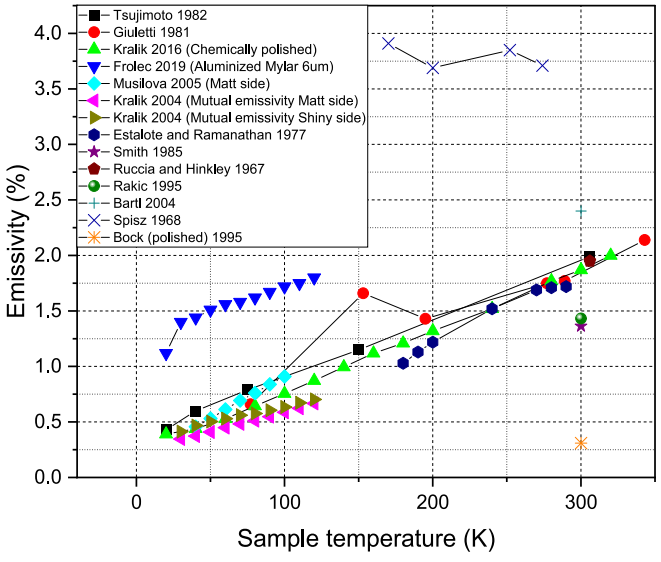


Fig. 36. Emissivity of aluminum at different sample temperatures.

specification of the material. Emissivity and absorptivity of stainless steel unfinished and finely turned are shown in Fig. 38 and found not to differ more than 15 % in their values, and in the lower temperature range, it's almost the same.

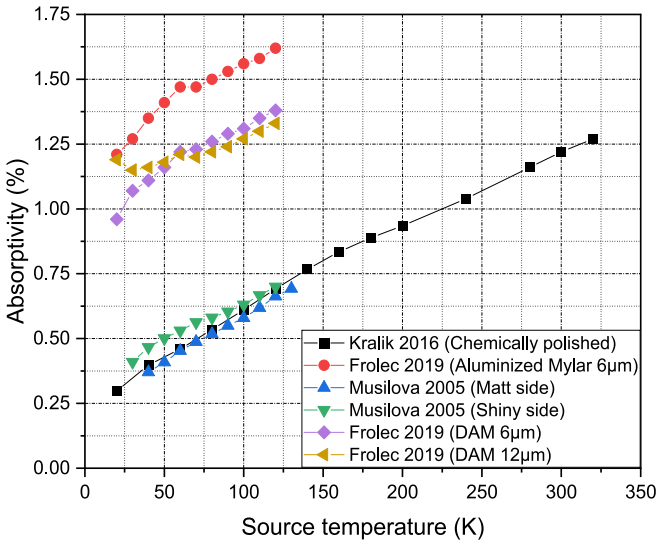


Fig.37. Absorptivity of aluminum at different source temperatures.

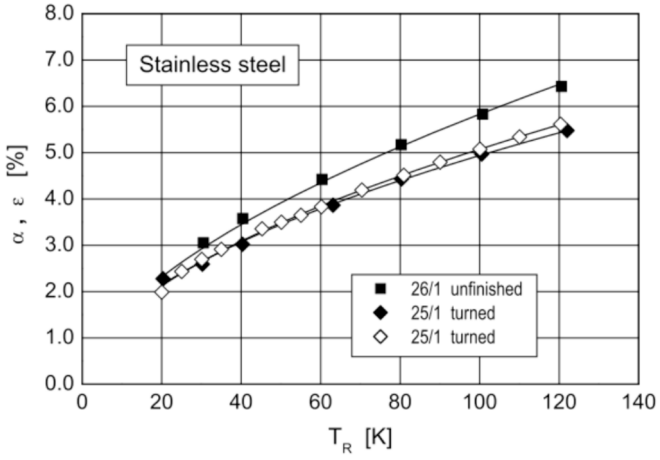


Fig. 38. Total hemispherical absorptivity (represented by solid symbols) and emissivity (represented by open symbols) of stainless steel (US type 304) 25/1 (finely turned on a lathe, 14 days exposed to air in the laboratory) and on 26/1 (Unfinished) [55].

Woods et al. [81] used ACR to measure the emissivity of polished 304L stainless steel within the temperature ranges from 80 K to 300 K and compared with the data from Roger et al. [99], which uses the calorimetric method with a temperature range of 340–1100 K and provides hemispherical emissivity rather than average emissivity as measured by Woods et al. The temperature gradient in emissivity for Woods et al. data agrees well with that of Roger et al as shown in Fig. 39.

Using a previously developed optical method by ACR to measure the emissivity of samples at 80 K, Woods et al. [28] calculated the emissivity of silver-coated SS304L Stainless steel and compared with the bare polished stainless-steel plate. The difference between the two samples is the silver coating of 5 μm on sample B in the last stage. The results of sample B have the lowest value of 0.0035 (80 K), which is approximately tenfold less than sample A, as graphically represented in Fig. 40.

Ma et al. [82] measured the spectral emittance of Quartz and stainless steel using FTIR. The results found that the spectral emissivity of the stainless-steel sample varied from 0.11 to 0.15, and it decreased with increasing wavelength. The spectral emissivity curve in Fig. 41 showed a short peak in the wavelength range of 8 – 10.5 μm, similar to Xu et al.'s findings [100]. The emissivity value of the stainless-steel sample in the 8 – 14 μm range was close to the results reported by Makino et al. [101],

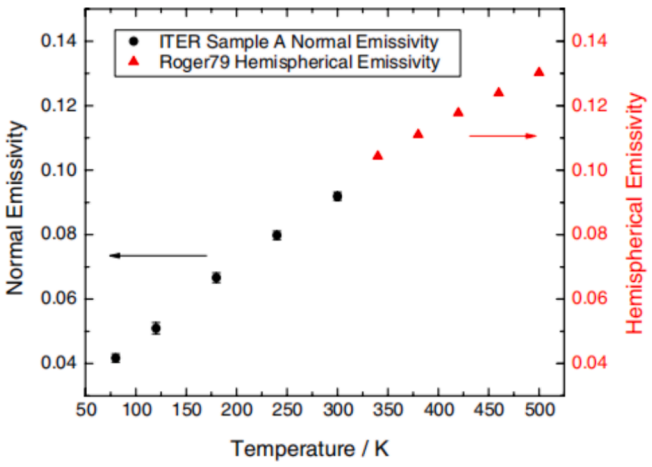


Fig. 39. Emissivity versus temperature (ranges from 80 K to 500 K) [81].

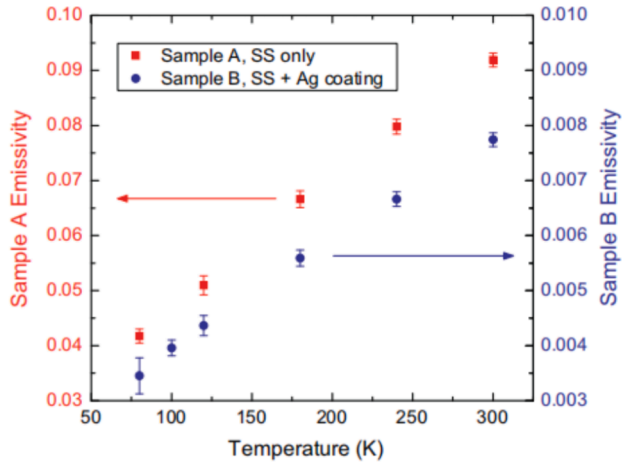


Fig. 40. Emissivity vs temperature for samples A and B, reported with error bars [28].

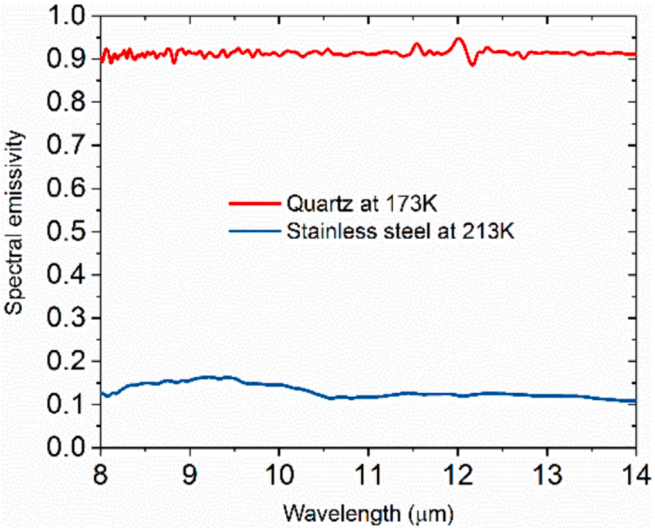


Fig. 41. Spectral emissivity of Quartz and stainless steel [82].

even with slight temperature differences. These findings suggest that the apparatus used in the study can accurately measure the spectral emissivity of materials with different emissivities.

Frolec et al. [14] used an instrument designed by Kralik et al. [55] to calculate the total hemispherical emissivity and absorptivity of metallic surfaces, specifically stainless steel. The surface finish minimizes the effect at lower temperatures but becomes prominent at higher temperatures. However, it also suggests that the impact of surface finish diminishes after ten years, as shown in Fig. 42.

Fig. 43 discusses the impact of ageing and surface treatment on stainless steel. The surface treatment involves sandblasting and fine turning. The figure shows that sandblasting causes a significant increase in total hemispherical emissivity. However, ageing does not affect the emissivity of the sample.

Figs. 44 and 45 provide a detailed analysis of the consolidated stainless steel's emissivity and absorptivity, respectively. Emissivity is the measure of a material's ability to emit radiation, while absorptivity is the measure of a material's capacity to absorb radiation. Both figures summarize the performance of stainless steel in terms of these properties. The data presented can be used to make informed decisions regarding using stainless steel in various industrial applications.

3.3. Superconductors

To study the properties of superconductors at cryogenic temperatures, several advanced experimental methods have been developed. One notable technique involves using a cryogenic bathysphere, an inverted Dewar flask submerged in cryogenic fluids, to measure the resistance of superconductors like Niobium Titanium (NbTi) and $YBa_2Cu_3O_{7-\delta}$ across a temperature range of 4 – 300 K. This method allows rapid temperature variation with minimal thermal lag, making it suitable for high-field applications [102]. Additionally, an advanced AC calorimetric technique based on optical AC calorimetry is used to measure thermal diffusivity. This method effectively detects superconducting transitions in materials like Hg-1223, utilizing various approaches such as frequency variation and the constant wave number method [103]. A cryogen-free superconducting magnet, cooled using a cryocooler, facilitates the measurement of electromagnetic properties at cryogenic temperatures without the need for liquid cryogens, providing long-term stability [104]. For microwave resonator measurements at millikelvin temperatures, a single-port calibration method is employed, enhancing accuracy by accounting for impedance mismatches in cryogenic systems [105]. Furthermore, the thermal expansion of superconducting wires, such as NbTi/Cu and Bi2212/Ag, is measured using a tensile method in a variable temperature cryostat, providing crucial data on material behavior under cryogenic conditions [106].

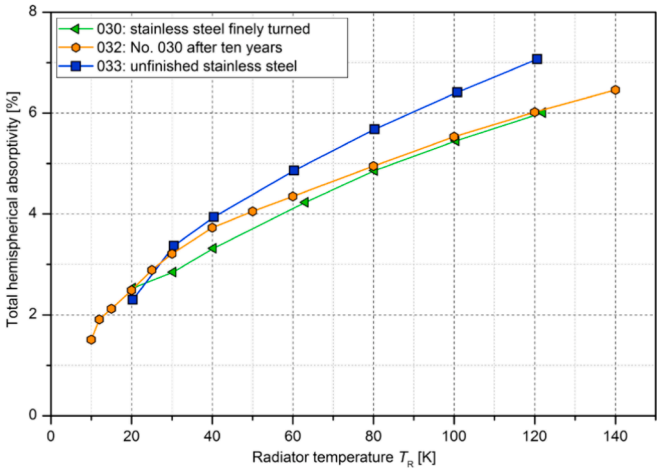


Fig. 42. Absorptivity of stainless steel varies with radiator temperature.

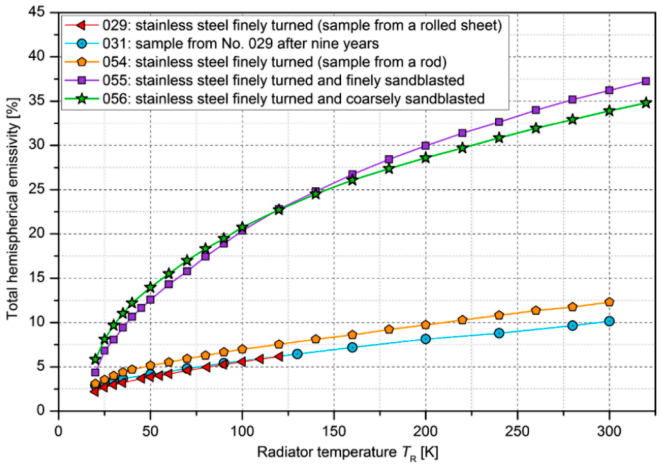


Fig. 43. Emissivity of stainless-steel samples with surface treatments.

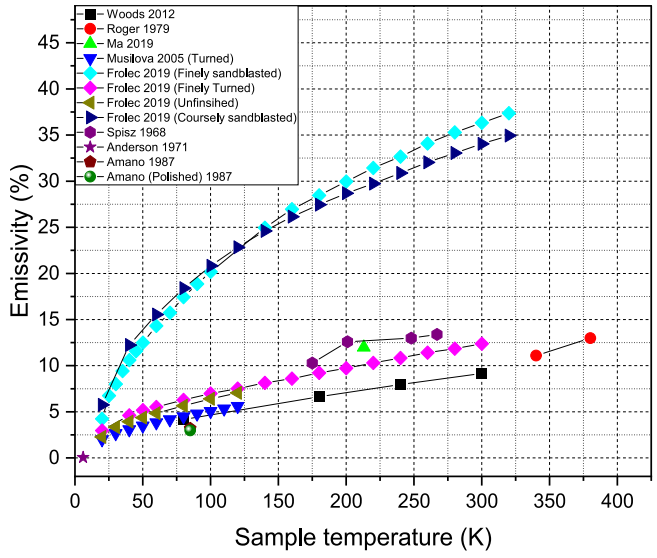


Fig. 44. Emissivity of stainless steel at different sample temperatures.

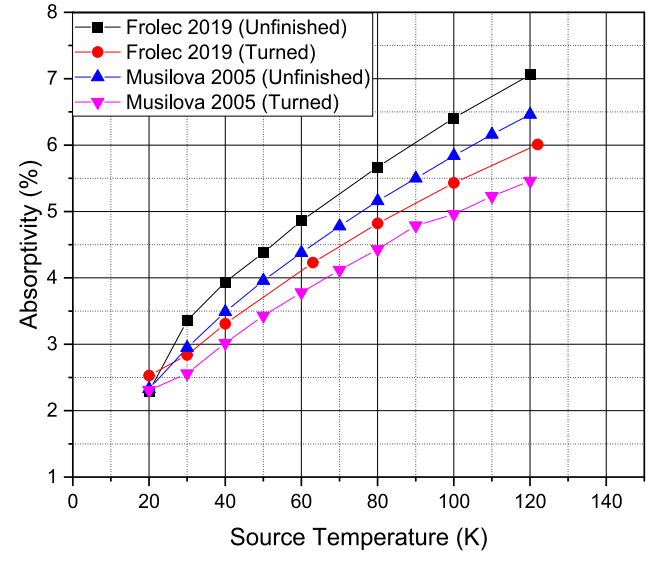


Fig. 45. Absorptivity of stainless steel at different source temperatures.

4. Future development

Future developments in emissivity/absorptivity measurement at cryogenic temperatures are expected to focus on several key areas:

- **Refinement of Measurement Techniques:** The field is likely to see improvements in existing emissivity measurement methodologies. This includes the development of cryogen-free apparatuses [58] and calorimetric methods to overcome the challenges posed by the weak radiative power of cryogenic radiators.
- **Advanced Materials:** An increasing emphasis is on exploring and optimizing new materials, such as low-emissivity thin films and coatings [63], designed specifically for cryogenic conditions. These materials ensure accurate measurements and maintain high performance at extremely low temperatures.
- **Innovative Instruments:** Future advancements will likely involve integrating high-precision instruments with enhanced capabilities, such as upgraded emissometers incorporating software enhancements and low-temperature blackbody references, to provide more accurate and reliable emissivity data at cryogenic temperatures.
- **Application in Emerging Technologies:** The progress in emissivity measurement will be crucial for applications in superconducting systems [28], space exploration, and other high-tech fields that depend on precise thermal management at cryogenic temperatures [24–26]. These advancements are expected to significantly enhance the stability and efficiency of technologies operating under cryogenic conditions.

5. Conclusion

In conclusion, the paramount role of emissivity in radiation heat transfer, particularly at cryogenic temperatures, and its application across various fields, such as space and planetary exploration, medical applications, instrument design, cryogenic storage and transportation was underscored in our review. Critical aspects of emissivity at cryogenic temperatures, the factors influencing it, and the prevalent methods for its measurement, such as calorimetric and radiometric techniques, were thoroughly examined. It was reported that traditional calorimetric methods, which are mostly steady state, are inadequate in capturing the dynamic nature of emissivity under certain cryogenic conditions. The literature significantly highlighted the need to develop transient measurement approaches that can provide accurate emissivity values during the dynamic cooling or heating phases of materials, albeit with the introduction of larger uncertainties compared to steady-state methods [107].

Moreover, the challenge in calibrating radiometric devices for cryogenic applications, requiring enhancement of calibration techniques to reduce the substantial uncertainties involved, was discussed [108]. The dependency of emissivity on the surface treatment of materials such as aluminum and stainless steel, which are widely used in cryogenic operations, was also covered. A summary of published data concerning the emissivity and absorptivity of these materials, offering valuable insights for ongoing research, was provided.

This review emphasized the importance of understanding emissivity in the context of cryogenic temperatures. It acted as a foundation for future research efforts to fill the identified gaps in measurement accuracy and calibration methodologies. Promoting advancement in these areas will significantly benefit the field, improving the reliability and precision of emissivity determinations in cryogenic settings.

CRediT authorship contribution statement

Abdul Rehman Hashmi: Writing – original draft. **Bo Wang:** Visualization, Validation, Supervision. **Fatima Saadat:** Writing – review & editing. **Zhihua Gan:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgement

This research was supported by the Baima Lake Laboratory Joint Funds of the Zhejiang Provincial Natural Science Foundation of China under Grant No. LBMHZ24E060004.

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